REVIEW ARTICLE

A Comprehensive Review of EEG Based Emotion Recognition: Theories, Methods, Evaluation, and Emerging Trends

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ABSTRACT

EEG based emotion recognition is a relatively new topic in the field of affective computing, which can enhance the ability of human–computer interaction and be used in the field of health care and adaptive technologies. EEG records the dynamics of emotions in the brain and offers information about emotions, which is difficult to obtain by using facial expressions and voice data. Even though there is considerable progress made in this field, comparing different studies is difficult due to differences in experiments, representation of signals, machine learning approaches, and evaluations. While some recent reviews focused on different aspects of EEG based emotion recognition, the relation between methods and generalization capabilities of models needs to be investigated more deeply. In this review, we cover the field from the neurophysiological bases of emotion recognition and emotion elicitation techniques to EEG recording, signal processing, feature extraction, and classification. We focus on the influence of all the mentioned factors on the reliability and generalization of models. In addition to traditional and deep learning methods, we cover data augmentation, transfer learning, and domain adaptation approaches to overcome the problem of limited amount of data and inter subject variability.

1 | Introduction

Emotion recognition is very important in the field of human behavior as the emotions influence cognition, decisions, and social behavior. Recognition of the emotions automatically allows intelligent machines to respond to the emotions of the user particularly in human-computer interactions [1]. The use of EEG technology in BCI provides further applications related to emotion recognition through the measurement of the brain's neural states [2]. For instance, using EEG in clinical studies, the functional connectivity changes due to depression have been studied [3].

EEG is a noninvasive technique used to record neural activity in the form of electrical potentials with very precise timing, thus making it possible to study emotions-related brain dynamics. While behavioral expressions can be modified by the person in order to avoid the revealment of emotional states, information provided by EEG is physiological and thus less manipulable than behavioral cues [4]. Still, EEG does not provide for an accurate detection of emotions, and multimodal approach might be useful in this case [1].

Despite all the benefits, there are challenges posed by EEG emotion recognition owing to the issues with artifacts, positioning of electrodes, conditions under which recordings are performed, and intersubject variability. Challenges with reproducibility also come from inconsistencies between different researches with regards to protocol design, nature of stimuli, methods of labeling the data, and preprocessing [5]. Data used in experiments is typically taken from well-known databases such as DEAP, SEED, and DREAMER [6].

In recent years, there have been developments that have included traditional machine learning techniques employing manually designed features and deep learning techniques enabling the learning of discriminative EEG representations. CNNs have been applied to learn features in EEG data [7], RNNs to learn the temporal dependency structure in EEG data [8], and GNNs to account for the channel connectivity [9]. Despite several reviews that have considered these models as well as the subject independent recognition aspect, issues of model generalization, differences in datasets used, and variations in the evaluations have not been addressed [10]. In view of this, the joint study of EEG data pre-processing, representation learning, classification approaches, and evaluation procedures would be useful.

Past literature reviews have improved the understanding of emotion recognition via electroencephalogram (EEG) signals through their investigation into signal processing, features, learning, transfer learning, and experiments. Nevertheless, these factors tend to be studied independently without considering the interactions among them in relation to the neurophysiological properties, methodologies, and experiments used and what impact they have on the performance and generalizability of the EEG based emotion recognition process. To this end, this review integrates these aspects to explore the EEG based emotion recognition problem from a holistic perspective. The contributions of this review include the following:

1. Connecting the neurophysiological bases to the methodology choices made: In this literature review, emotion theories, stimulus design, recording of EEG signals, and brain activations are linked to the later stages of recognition. The study of such connections reveals how the experimental setup and properties of recorded neural signals impact signal processing and modeling.
2. Formulation of an integrated workflow centric framework for EEG emotion recognition: There are different frameworks that exist, each classified according to the various levels involved in emotion recognition such as pre processing, feature extraction/selection, modeling and evaluation. The dependencies of these stages are analyzed and described in order to show how decisions taken at one stage affect other stages.
3. Synthesis of relationship between EEG representation and learning approaches: In this review, it is considered how temporal, spectral, spatial, and time frequency representations are used in classical machine learning approaches and neural networks such as convolutional, recurrent, graph based, attention based, and hybrid networks. The approaches depend on the type of information being used, their dependence on features of brain regions and frequency bands, and on their computational demand.
4. Connecting key methodological problems with their solutions: The current review integrates the problem of feature redundancy, overfitting, lack of training data, and variability of subjects into one single list which is accompanied by the corresponding solution methods. The methods of feature selection, regularization, data augmentation, transfer learning, domain adaptation, domain generalization, and fine tuning are evaluated against their purpose of improving recognition accuracy.
5. The requirements for applying experimental achievements in practice: Apart from classification accuracy, the analysis includes robustness, computational cost, channels needed, variability of recordings, and generalization on novel subjects. All these aspects define the research directions towards making EEG based emotion recognition more robust and applicable in practice.

As stated in Table 1, the unique value of this review is based on an integrative examination of neurophysiological background, recognition processes, approaches to learning, evaluation techniques, and implementation issues. Such an approach allows for exploring connections between these aspects and for defining limitations which have not been fully addressed yet.

2 | Theoretical and Methodological Foundations

The theoretical background defines EEG-based emotion recognition as a framework that links measurable neurophysiological activity to computational representations of affect. It models emotional state as a structured pattern distributed across time, frequency bands, and scalp locations, and it treats learning as the problem of extracting robust features or representations from noisy, non stationary brain signals. The paper also shows two methods for using emotion labels including the following paradigms: the discrete classification paradigm in which EEG segments are classified into separate emotion states and the continuous modeling paradigm in which the prediction of valence and arousal is performed [4]. These paradigms shape methodologies of design, evaluations and generalization of subjects and sessions [16].

2.1 | Literature Search Strategy

Targeted literature identification and screening methods were used to conduct the analysis on the research work performed for emotion recognition using EEG signals. This method focused on the scientific and methodology importance and comparative value of the empirical results found, following existing methods for conducting a literature review.

2.1.1 | Databases and Source Identification

The literature used in this study includes the collection of research articles, review articles, and their respective references. Literature in this study covers the searches performed in various academic search engines including IEEE Xplore, ScienceDirect, Web of Science, PubMed, and Google Scholar. Studies were selected in various areas such as EEG recording, preprocessing of EEG signals, features extraction, emotion classification, deep learning, and generalization across subjects. These databases represent sources documented in the reviewed literature rather than an independently recorded systematic search conducted for the present review.

TABLE 1 | Comparison of key research dimensions covered by recent reviews on EEG based emotion recognition.

Research Dimension	Jafari et al. 2023 [11]	Vempati and Sharma 2023 [12]	Erat et al. 2024 [13]	Wang et al. 2024 [14]	Ma et al. 2024 [4]	Geng et al. 2025 [10]	Chen et al. 2025 [15]	This Work
Emotion Theory	–	✓	✓	–	–	✓	✓	✓
Stimulus Design	–	–	✓	–	–	–	–	✓
EEG Acquisition	–	✓	✓	–	–	✓	✓	✓
Multimodal Modeling	–	–	–	✓	–	–	–	✓
Feature Fusion	–	–	–	✓	–	–	–	✓
Data Augmentation	–	–	–	✓	–	–	–	✓
Model Taxonomy	✓	✓	✓	–	✓	✓	✓	✓
Domain Adaptation	–	–	–	✓	✓	✓	–	✓
Subject-Independent Evaluation	–	–	–	✓	✓	✓	✓	✓
Evaluation Reproducibility	–	–	–	–	–	–	–	✓

Note: ✓ indicates substantial coverage of the specified research dimension, whereas indicates that the dimension is not treated as a distinct major focus. The comparison reflects the thematic emphasis of the reviewed articles rather than their scientific quality.

2.1.2 | Search Keywords

The keywords for the literature screening include EEG, electroencephalography, emotion recognition, and emotion classification. Other keywords include EEG feature extraction, signal preprocessing, machine learning, deep learning, subject-dependent recognition, subject-independent recognition, cross-subject classification, transfer learning, domain adaptation, and data augmentation. The search criteria are formulated based on the Boolean expression (EEG OR electroencephalography) AND (emotion recognition OR emotion classification). Additional terms are used to capture papers dealing with certain challenges of the field. This approach was used in similar reviews on EEG emotion recognition.

2.1.3 | Inclusion and Exclusion Criteria

Studies were considered eligible when they directly investigated EEG based emotion recognition or provided relevant methodological insights into emotion elicitation, EEG acquisition, preprocessing, feature extraction, classification, or performance evaluation. Those studies which focused on cross disciplinary variability, generalization, transfer learning, domain adaptation, and data augmentation were also considered relevant. Original studies that included experimental designs that can be recognized and interpreted were preferred over review articles, which were retained to give a theoretical background.

The papers that had nothing to do with emotion recognition or had insufficient information about methodology were discarded. Duplicate publications were not treated as independent evidence, and unverifiable performance claims were not included in quantitative comparisons. The selected literature was subsequently organized according to its methodological focus, experimental settings, learning approaches, and evaluation protocols to support a critical synthesis of developments and remaining challenges in EEG-based emotion recognition.

2.2 | Emotion Recognition

Emotion recognition is a crucial objective of affective computing, allows for customized and emotion sensitive human computer interaction (HCI). The need for such technology has increased due to the increasing requirement for personalized and emotion aware human computer interaction (HCI) because of the possibility of improving the interaction of computer systems with users by detecting their affective state [1]. The traditional techniques depend upon facial expressions, speech, gestures, and text, which can be affected by environment and are consciously hidden by people at times. In contrast, the physiological measurements like electroencephalography (EEG) give an indication of brain activity related to emotion.

Emotion detection using EEG works with emotions detected based on neural rhythm and spatial activities. Conventional techniques use preprocessing, temporal, spectral, and time-frequency features extraction, then applying the learning algorithm for the classification of the extracted features. Meanwhile, deep learning is able to extract complex spatiotemporal features automatically without the need for extensive manual feature extraction. Nevertheless, subject-to-subject variability, the recording environment, and the process of elicitation of emotions make its application limited in the field. [17].

2.3 | Neurophysiological Foundations of Emotional States

Neurophysiological accounts of emotion in EEG emotion recognition interpret emotional episodes as changes in distributed brain activity that manifest as time varying and band limited oscillatory patterns across channels. EEG acquisition during affective stimulation, therefore, produces signals with spatial structure across electrode locations, temporal structure due to stimulus duration and sliding window segmentation, and frequency structure because analysis commonly separates δ, ϑ, α, β, and γ rhythms, each of which exhibits different responsiveness under affective conditions.

In this context, EEG based emotion recognition pipelines generally follow either a conventional sequence of denoising, feature extraction, and classifier training or an end to end learning approach. Both approaches depend on the availability of labeled data and the extent to which training and test samples satisfy the assumption of being independently and identically distributed (IID) [16]. When these assumptions fail especially under cross

subject, cross session, or cross database settings emotion related neural patterns appear heterogeneous, motivating modeling strategies that explicitly target domain gaps and subject bias. EEG emotion recognition surveys formalize feature construction as multi-domain representation learning. Temporal features describe transitory to long lasting changes in events, while spectral features include anything like Bandpass filtering, Fourier transformation, and other transformations like them. Spatial features describe interchannel relationships that correspond to the spatial organization of the cortical structure of the human brain and its variability. This multidomain stance links directly to contemporary architectures (CNN/RNN/GNN/Transformer variants) that operationalize spatial, temporal and spectral fusion and to transfer learning approaches that seek domain invariant representations under strong inter individual variability.

Benchmark datasets encode these neurophysiological assumptions by pairing affective stimuli like, videos or music with EEG recordings and either discrete labels or continuous ratings, thereby defining the target emotional state space and the evaluation protocol.

2.3.1 | Discrete Emotion Representation Models

Discrete emotion theories categorize emotions in a class form that starts from the general positive, neutral, and negative emotions to discrete emotions like happiness, sadness, and anger. Classification in EEG emotion recognition is concerned with the separation of the emotional classes using the neural activities captured as EEG data over time, frequency, and electrodes.

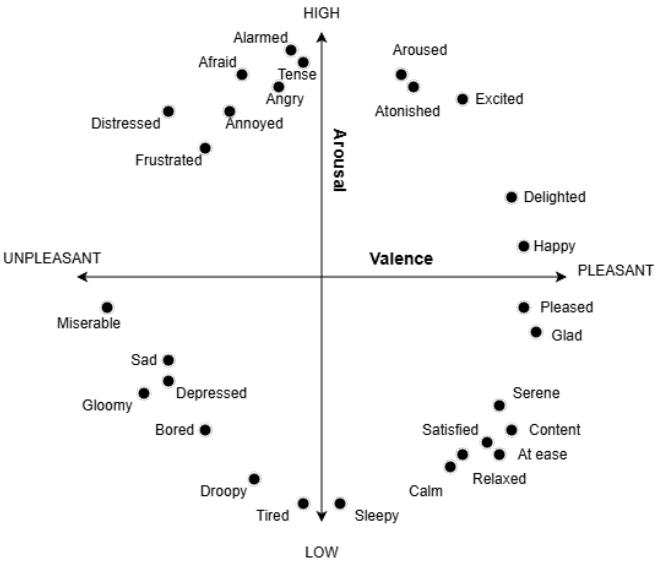


FIGURE 1 | Conceptual representation of affective states in the two-dimensional valence-arousal space.

Commonly used features are PSDs and DEs in different frequency bands and the spatial information of multiple electrodes. Conventional classifiers such as SVM and KNN use manually designed features, while deep learning methods automatically extract the EEG features. Spatial and spectral features can be learned using CNNs, while temporal features can be extracted using RNNs. Relations between different electrodes can be represented by GNNs, while attention and hybrid models combine these abilities [9].

Fig. 1 illustrates the valence-arousal space, in which emotional states are represented by their relative positions rather than isolated labels. Continuous ratings may also be divided into binary high/low classes or four valence–arousal quadrants, although these transformations discard some information about emotional intensity.

The recognition accuracy is also affected by the evaluation environment. While subject-dependent systems can take advantage of the patterns specific to an individual, subject-independent and cross-dataset classification needs to account for the variation in the EEG distributions. These changes can be overcome via transfer learning and domain adaptation, which involve the transfer of knowledge via instances, features, parameters, or relations. This implies that the usefulness of discrete emotion models is not limited to their classification performance alone.

2.3.2 | Continuous Dimensional Emotion Representation Models

The dimensional continuous model describes emotions in terms of dimensions, including valence (pleasantness) and arousal (activation). Dominance is also sometimes considered as a dimension. In contrast to the discrete emotion classification, the dimensional models permit estimating continuous affective ratings or categorize them into groups, for instance, high and low valence and arousal [15, 18].

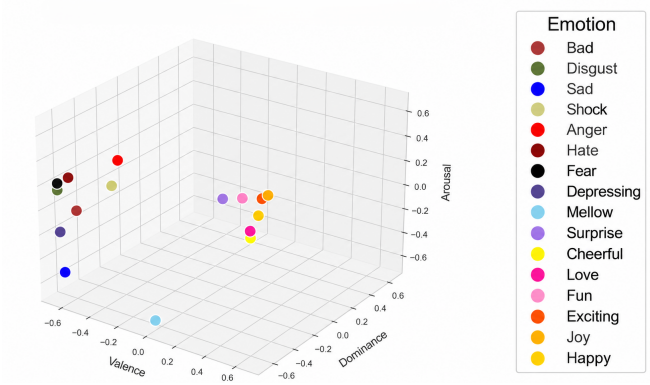


FIGURE 2 | Three dimensional representation of selected emotional states in the valence–arousal–dominance (VAD) space, illustrating their distribution across emotional pleasantness, activation, and perceived control.

Dimensional models are related to the EEG-based emotion recognition because the EEG is an indicator of time-varying neural processes involved in emotional responses [19]. The EEG signal is sensitive to various types of noise including physiological ones. This is the reason why preprocessing is usually done before any analysis or feature extraction from the signal [20]. After that, depending on the recognition method, EEG signals can be segmented into windows to get PSD, DE, or other features or can be analyzed by means of deep learning architecture which learns spatial and temporal information [18, 21].

The three dimensional emotion model extends the traditional valence-arousal framework by introducing dominance as an additional dimension of emotional experience. Valence describes whether an emotion is pleasant or unpleasant, arousal indicates its level of activation, and dominance reflects the degree of control an individual perceives over the emotional experience. As illustrated in Fig. 2, emotional states can be represented as distinct points within this three dimensional space, providing a more detailed description of their affective characteristics. By considering these dimensions together, the model offers a broader representation of emotional experiences, which can be useful for interpreting EEG responses and developing emotion recognition systems that account for variations in affective states.

A persistent limitation is the variability of EEG distributions and affective ratings across subjects, sessions, and datasets. Negative transfer is one possibility of distribution mismatches in transfer learning and domain adaptation that might hinder the process of generalization. Thus, successful dimensional modeling should involve reliable labels and proper preprocessing as well as evaluation processes of the model.

2.4 | Emotion Induction and Stimulus Design

The induction method affects the way in which the affective state is induced, labeled, and compared to the EEG signal. Visual, auditory, and audiovisual stimulation have been extensively utilized in experiments of EEG-based emotion recognition, mostly based on clearly specified presentation intervals to construct experimental trials out of the obtained EEG data [22, 23]. Post-stimulus questionnaires are helpful in making a link between EEG recording and the experienced emotion by the subject [11].

Tasks and memories can trigger relevant emotions. However, the unpredictability and subjective nature of the stimuli make it hard to control the experiment. The choice of stimuli for an experiment should combine standardization, relevance, and ecological validity. When working on multimodal data, synchronizing readings from peripheral channels to EEG recording is important in order to identify the same response [24].

The differences in stimulus type and emotional induction procedure might create distributional shift between corpora, which would impair the ability of the model to generalize on different corpora [25]. The use of the new contrastive and self-supervised paradigms, on the other hand, further motivates designing experiments that allow for acquiring diverse and comparable EEG recordings for building robust affective representations [26].

2.5 | Progress in Data Acquisition Technologies

Improvements in acquisition of EEG data have allowed emotion recognition to move from laboratory settings towards wearable devices and mobile applications. Traditional multichannel EEGs can cover large areas of the scalp but need extensive preparations and limit the movement of subjects while wearable EEG systems are more convenient. Therefore, the choice of the device should take into account the quality of data, comfort of participants, and conditions of use, since the differences in acquisition methods may affect the performance of models [2].

The number and spatial arrangement of EEG electrodes determine scalp coverage and influence the spatial information available for emotion recognition. Fig. 3 shows a multichannel configuration covering frontal, central, temporal, parietal, and occipital regions of the brain. Adaptive graph construction enables models to capture emotion-related relationships among EEG channels, while sparse connectivity representations can reduce graph-processing complexity [27].

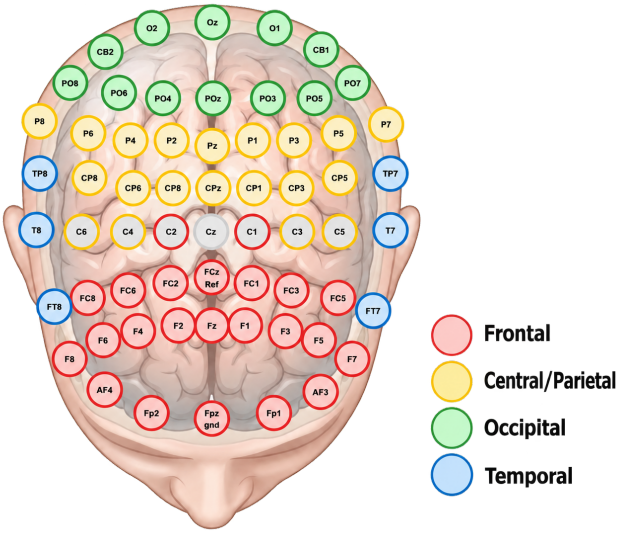


FIGURE 3 | Illustrative multichannel EEG electrode placement across major scalp regions.

TABLE 2 | Representative EEG frequency-band findings across scalp regions and intracranial structures during emotion-related tasks.

Region	Reported oscillatory findings
Frontal	Alpha (8–13 Hz) and frontal-midline theta (2–7 Hz, study-defined), examined for emotional valence and arousal prediction [28]
Temporal	Beta (14–30 Hz) and gamma (31–50 Hz) features over lateral temporal scalp regions during emotion classification [29]
Parietal	Alpha (8–12 Hz) and mu (10–12 Hz) modulation in source-localized parietal regions during emotional stimulus processing [30]
Occipital	Alpha (approximately 8–12 Hz) desynchronization over posterior, including occipitotemporal, sensors during emotional picture viewing [31]
Central (sensorimotor)	Mu (8–13 Hz) desynchronization during emotional-face perception [32]
Limbic-related (intracranial)	Amygdala theta phase (3–12 Hz) coupled with hippocampal gamma amplitude (50–75 Hz) during aversive memory formation [33]

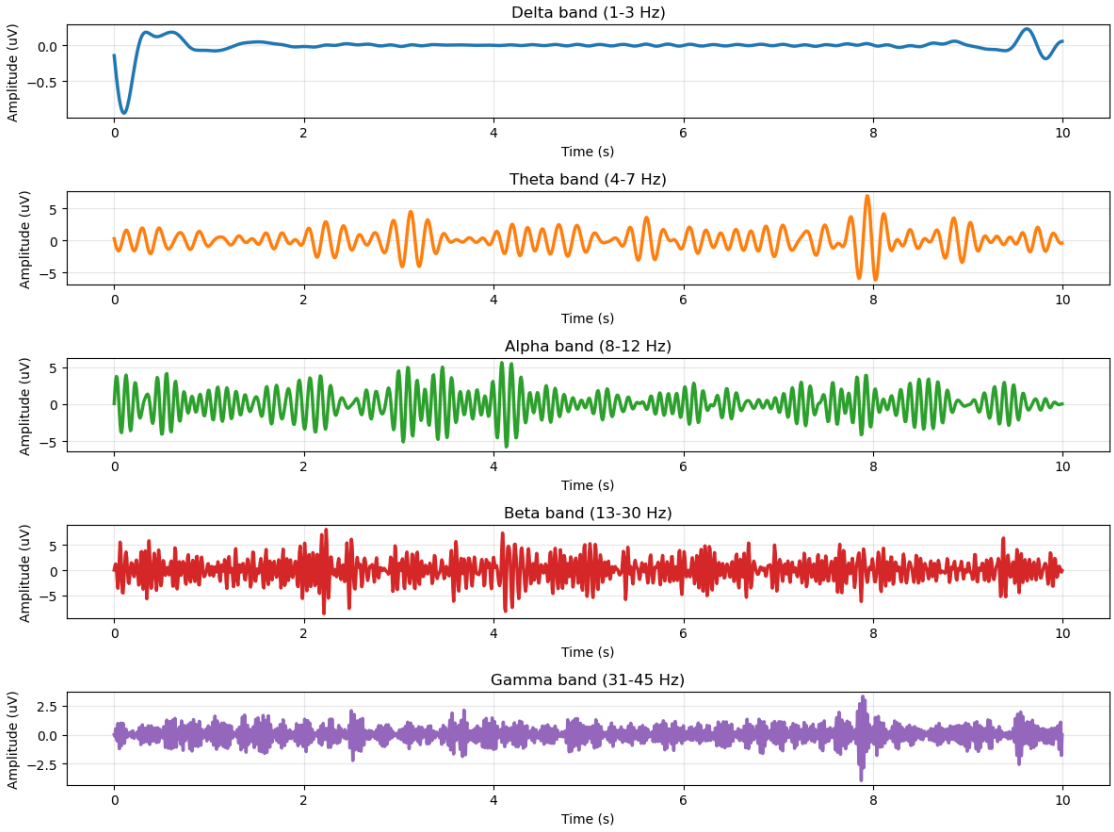


FIGURE 4 | Illustrative EEG waveforms across delta (1–3 Hz), theta (4–7 Hz), alpha (8–12 Hz), beta (13–30 Hz), and gamma (31–45 Hz) frequency bands.

Improvements in wireless and wearable EEG technology have increased possibilities to record brain activity outside traditional lab environments. Low-cost and low-channel EEG equipment may increase availability; however, insufficient spatial resolution, motion artifacts, and inconsistency in electrode-skin contact might influence the quality of recordings [34]. Real-time EEG analysis may also provide new opportunities for mobile affective applications. But the differences in EEG recording devices and conditions should be taken into account while assessing generalization of models across devices.

EEG is traditionally processed within five frequency bands: delta, theta, alpha, beta, and gamma. They allow analyzing brain activity related to emotions from different perspectives; however, the limits of these bands may differ depending on the research [35, 22]. EEG oscillatory activity varies across scalp regions and experimental conditions. Table 2 summarizes representative frequency-band findings associated with emotional processing. These associations are illustrative rather than exclusive regional characteristics, and the limbic-related findings are based on intracranial recordings, while the frequency band's oscillations are shown in Fig. 4. It is worth noting that these are not specific for some regions or particular emotions of the brain but depend on where they are recorded.

2.6 | Multimodal Emotion Modeling

Emotion modeling in a multimodal framework makes use of behavioral features including facial expression, speech, textual inputs, and gestures along with physiological features including EEG, ECG, EMG, electrodermal activity, and respiration. Different features cover different facets of affective experience and could lessen the reliance on one feature which is inherently prone to noise, ambiguity, or context dependency [36, 37].

In EEG-based emotion recognition, neural measurements provide temporal insights into neural activity, whereas behavioral and physiological measurements provide insights into the expressive or autonomic response. Taken together, these techniques, with their variety in terms of both the timing and type of measurements, could help to gain another view on emotions. The modalities of behavioral and physiological measurements are presented in Fig. 5.

Multimodal fusion is usually classified into early, intermediate, and late stages of fusion [37]. Early fusion fuses features prior to classification, thus making possible for interaction between modalities but increases dimensions and sensitivity to misalignment [37]. Intermediate fusion provides joint representation, which is able to represent more complicated dependencies but requires higher model complexity [38]. Late fusion merges output of classifiers for individual modalities, providing modularity but ignoring feature dependencies [38]. Attention and gating strategies allow for adaptation of the influence of each modality, but the efficiency is conditioned by the quality of the data [39, 40].

Multimodal datasets like DEAP and AMIGOS help in studying complementary affective signals [36, 41]. Nevertheless, additional modalities do not guarantee better generalization: synchronization errors, missing signals, inter-subject variability, and computational overhead remain important challenges. Reliable multimodal EEG systems therefore require carefully designed

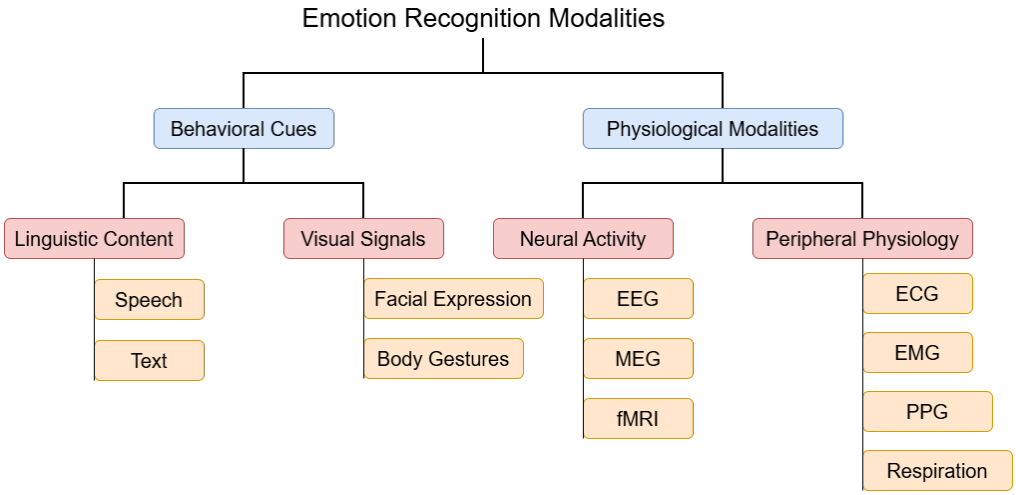


FIGURE 5 | Classification of emotion recognition modalities into behavioral cues and physiological signals, highlighting linguistic, visual, neural, and peripheral physiological measurements.

fusion strategies and evaluation against comparable unimodal baselines.

2.7 | EEG Correlates of Emotional States

Emotion-related EEG activity takes the form of patterns present in frequency bands, scalp areas, and time, rather than being confined to the amplitude of one particular wave or the rhythm of the brain. The above consideration supports the use of multi-frequency and spatial approaches to describe the EEG response related to different emotions. For example, Chen et al. used differential entropy in combination with spatial features extracted based on electrode distribution to create frequency-specific brain maps for emotion recognition [42].

Understanding these correlations is important since EEG oscillations are not unique identifiers of emotions experienced by an individual. The correlation of the two can be determined by frequency band, scalp area, and experimental setting. Functional connectivity studies in various frequency bands have found emotion-specific differences in delta, alpha, and beta frequencies, with contribution made by various brain areas [43]. Thus, the spectral properties need to be understood in the context of their spatial distribution rather than being considered separate identifiers of emotions.

Beyond local spectral characteristics, relationships between EEG channels provide another perspective on emotion-related brain activity. Choong et al. performed Pearson correlations on electrode-pair combinations within the frequency ranges of alpha, beta, and gamma waves in relation to emotional stimulation [44]. Such measures can complement channel-wise spectral features, although statistical correlations alone do not establish direct neural connectivity. Temporal information is also relevant, as emotion-related EEG patterns can be examined across successive windows during stimulus presentation. Liu et al. demonstrated this approach through real-time classification of movie-induced emotions using overlapping EEG segments [45]. Similarly, Qing et al. introduced emotional activation curves to characterize the progression of estimated emotional states throughout experimental trials [46].

Overall, the EEG correlates of emotional states are better characterized through complementary spectral, spatial, inter-channel, and temporal information than through individual frequency bands alone. These representations provide a basis for feature selection and model design, but their consistency and neurophysiological interpretation require further assessment across subjects, recording sessions, and experimental conditions.

2.8 | EEG Signal Processing

The objective of EEG signal processing in emotion recognition is to transform the raw and noisy EEG signals into representations that are invariant to inter subject and inter session variability, while capturing the neurophysiological dynamics that are relevant to emotion [47]. In this regard, the signal processing pipeline is equivalent to a series of conditioning steps that include: (i) suppression of non cerebral noises, (ii) standardization of the temporal structure to facilitate learning, and (iii) definition of feature spaces that reveal the temporal, spectral, and spatial characteristics of emotional states. The signal processing pipeline is not just a pre processing step, it is a critical component that determines the information that can be extracted and utilized by the models, and it affects the robustness, interpretability, and generalizability of the models directly [7].

Affective EEG is usually measured using a specific protocol in which the emotional response is dynamic over time. Therefore, the processing pipeline must accommodate the ability to represent transient responses like, early perceptual responses and sustained responses such as, sustained affective responses. As the datasets move away from the controlled environment, the pipeline focuses more on the resilience of artifacts and stability across domains, encouraging the use of structured denoising, segmentation, and representations that align with the current deep learning encoder models [48].

The signals received through an electroencephalogram are known to be quite noisy and prone to a number of artifacts, and therefore preprocessing is an essential step for the recognition of emotions through EEG signals [49]. The preprocessing step is concerned with the cleaning and preprocessing of the raw EEG

signals to improve their quality and normalize the input signals for the subsequent steps of feature extraction and classification [13]. The preprocessing of EEG signals is an essential step for improving the SNR of EEG signals and removing external noise and artifacts such as power line noise and physiological artifacts such as eye blinks and muscle movements for better performance of the classification model for emotion recognition. It is thus important to apply proper filtering, artifact removal, normalization, and other cleaning procedures before the feature extraction step [50]. Moreover, the use of similar preprocessing procedures such as the use of similar filters and scaling helps to eliminate variability and errors in the comparison of signals from different domains [48]. Below, we discuss a variety of critical preprocessing procedures and approaches used in recent studies on EEG based emotion recognition.

2.8.1 | Filtering and Noise Reduction

Filtering and denoising play an important role in EEG preprocessing, where the objective is to suppress noise that does not originate from the brain and retain brain activity that can be used for further analysis. The recordings obtained through EEG are very sensitive to physiological noise and environmental noises. Consequently, appropriate preprocessing is important for distinguishing neural activity associated with cognitive and affective processes from unwanted interference [15].

Filtering is particularly relevant to EEG-based emotion recognition because emotion-related information can be represented through frequency-dependent neural activity. EEG signals are commonly analyzed across the delta (δ), theta (θ), alpha (α), beta (β), and gamma (γ) frequency bands. Correlations between EEG channels in terms of emotions were considered only for alpha, beta, and gamma bands [44]. But the recorded signal can be corrupted by eye movements, muscle activity (EMG), electrode artifacts, and power line noise, which might interfere with the actual brain signal [20].

Noise removal methods involve the application of band-pass filters to constrain the frequencies to be analyzed, notch filters to suppress narrow-band electrical noise when needed, and spatial filters or re-referencing to change the influence of the common signals. Other noise removal strategies are regression and blind source separation algorithms, such as independent component analysis (ICA). Wavelet decomposition can also be used together with ICA to decrease artifact contamination and preserve relevant neural information. The process of filter design requires special attention to frequency and phase responses; zero-phase Butterworth filtering has been used in the process of EEG noise reduction [51].

However, noise reduction is associated with a trade-off between artifact rejection and preservation of neural information. Too strict constraints of frequency responses may lead to the loss of EEG activity, while the complete removal of the components affected by artifacts may result in the loss of neural information. Thus, the parameters of preprocessing should be set based on specific features of recording and their evaluation in both temporal and frequency domains.

2.8.2 | Artifact Removal

Conventional frequency-selective filters cannot completely remove ocular and muscular artifacts when their spectral content overlaps with genuine neural activity. Eye blinks, eye movements, and facial muscle contractions can therefore contaminate EEG recordings and distort features used for emotion recognition. Ocular artifacts particularly affect frontal recordings, whereas muscle-related contamination can also influence electrodes near facial and temporal muscles [20].

2.8.2.1 | Independent Component Analysis (ICA). The wavelet technique is based on decomposing the EEG signal into coefficients that carry information at different scales in time and frequency. The coefficients that carry artifacts can be modified by thresholding and then inverse wavelet transform to get the corrected signal. Adaptive thresholding techniques have been used for removing eye blink artifacts from EEG signals while keeping the waveform of the EEG signal intact. In the work by Phadikar et al. [52], an automatic approach based on discrete wavelet decomposition with optimal coefficient thresholding has been suggested to remove the eye blink artifacts. However, the efficiency of wavelet-based artifact elimination is dependent on the choice of wavelets, decomposition level, and thresholds [51].

2.8.2.2 | Principal Component Analysis (PCA). Principal component analysis (PCA) performs orthogonal transformation to decompose the EEG measurements into a set of independent principal components, which is ordered by their variance. Principal components that are more related to noise can be discarded before reconstructing the EEG signals. Despite PCA being a simple method, it seeks to maximize variance and not independence.Consequently, neural and artifactual activities may remain mixed, particularly when their signal characteristics overlap. Removing components solely because they explain substantial variance may therefore result in the loss of relevant neural information [20].

2.8.2.3 | Wavelet-Based Artifact Removal. Wavelets provide solutions for decomposing EEG signals into coefficients that provide information at different time and frequency scales. These corrupted coefficients may undergo a process known as thresholding and then transformed back through inverse wavelet transformation to form the new corrected signal. For the elimination of ocular artifacts, several techniques of adaptive thresholding have been examined to attenuate the eye-blink artifacts while preserving the EEG signal. Phadikar et al. [52] designed an automatic method using discrete wavelet decomposition with optimum threshold values for removal of eye-blink artifacts. Nevertheless, the performance of wavelets for artifact correction is determined by the choice of the wavelet, decomposition level, and threshold value.

2.8.2.4 | Hybrid ICA–Wavelet Techniques. ICA-wavelet hybrid methods involve combining source separation with wavelet artifact reduction. ICA is used to decompose multichannel data into independent components, after which wavelets can reduce artifact coefficients in lieu of eliminating whole components. Such a process intends to preserve brain activity that could coexist with artifacts. Automated procedures have also made use of statistical features such as kurtosis and multiscale sample entropy

to detect artifact components prior to wavelet artifact reduction [50].

The relevance of hybrid preprocessing to EEG-based emotion recognition was investigated by Bigirimana et al., who compared conventional ICA with a hybrid ICA–wavelet approach. Their experiments showed improved emotion-classification accuracy with the hybrid approach when statistical and wavelet-based features were employed, under the evaluated experimental conditions [53]. Nevertheless, successful application depends on appropriate component identification, wavelet parameters, and preservation of relevant neural information. Computational requirements and possible distortion of reconstructed signals must also be considered.

In conclusion, there is no method for artifact removal that can ensure artifact elimination without losing any data. Thus, it is important to evaluate the effectiveness of pre-processing methods in terms of their capability to reduce artifacts and preserve emotion-related features of the EEG. In case of emotion recognition being the final goal, classification accuracy must also be taken into account.

2.8.3 | Channel Selection and Spatial Filtering

The two techniques of channel selection and spatial filtering can be viewed as complementary tools for enhancing the spatial resolution and processing efficiency of electroencephalogram data. The technique of channel selection tries to remove redundancy in the data through choosing informative electrode channels. Wang et al. demonstrated that normalized mutual information-based channel selection could substantially reduce the number of EEG channels while preserving most of the emotion recognition performance [54]. This is particularly relevant because emotion-related EEG information can be distributed across multiple scalp regions, with different electrodes contributing unequally to affective representations.

Identifying informative spatial regions is important for EEG-based affective computing, particularly when developing models intended to generalize across individuals. Attention-based analysis has identified relevant information in frontal, temporal, parietal, and occipital EEG channels. According to Valderrama and Sheoran, the electrodes positioned on the circumference of the head were assigned higher attention weights in a subject-independent emotion recognition framework [55]. This implies that spatial importance is not limited to only one area of the brain. Although reducing the number of electrodes can make it easier to acquire and process signals, learning the importance of channels does not mean the remaining channels are redundant.

Different approaches are available for representing and prioritizing spatial EEG information. Data-driven channel selection can identify electrode subsets using statistical relationships, whereas attention-based methods can learn spatial feature importance without explicitly removing channels. Wang et al. proposed a hierarchical Transformer model that captures spatial dependencies at electrode and brain-region levels [56]. Spatial filtering addresses a different aspect of EEG processing. Common-average referencing (CAR) subtracts the mean potential across the included electrodes from each channel, whereas surface Laplacian filtering estimates local spatial gradients and provides a reference independent representation that emphasizes more localized activity [57].

On the whole, there is a complementing pair of methods in the form of channel selection and spatial filtering that one may apply for dealing with the inherent spatial complexities of an EEG recording session. However, channel selection may end up losing valuable spatial features, and the application of spatial filtering may reduce broad-spectrum activity or even distort scalp potential maps. These two methods will depend largely upon electrode placement and recognition task.

Table 3 shows representative EEG preprocessing methods with various goals and measures used in evaluations. The goal of artifact suppression focuses on contamination in the acquired data, but the baseline correction considers differences between resting-state data and stimulus-evoked data acquisition. On the other hand, channel selection deals with input dimensionality reduction, whereas a real-time pipeline involves preprocessing and feature extraction subject to computation resource constraint. Hence, all methods have different goals and cannot replace each other.

The above observations reveal that the evaluation of preprocessing is not limited only to how clean the output is. If an algorithm effectively suppresses artifacts, it does not necessarily mean that the emotional classification improves, especially when neural information is being suppressed by this preprocessing step. Similarly, dimension reduction or computational efficiency cannot always ensure better generalization. In summary, preprocessing should be selected based on artifact types, baseline availability, channel configuration, and further recognition tasks.

2.8.4 | Feature Extraction

Feature extraction transforms the preprocessed data of the EEG into features that represent the brain activities associated with the emotions. Features can include the temporal dynamics, frequency brain oscillations, and the spatial connectivity between different electrodes. In classical approaches, the process of feature selection is done manually by using statistics and signal processing methods, while in deep learning approaches it is done automatically.

Feature extraction is especially important for the recognition of emotions from EEG due to the fact that the brain activities associated with emotions can vary depending on frequency, scalp location, and time. Thus, combining multiple sources of information can result in a more comprehensive representation of EEG activity compared to individual feature sets. Nevertheless, the usefulness of such representations is defined by their capacity to retain relevant information without redundancy.

Recent EEG emotion recognition methods increasingly employ multidimensional representations to capture relationships among channels and frequency bands. Li et al. [60] constructed three-dimensional EEG feature arrays from frequency-specific electrode maps and used dilated fully convolutional networks to learn cross-channel and cross-band relationships. In order to learn temporal dependencies, Lew et al. [61] explored spatial-temporal representations by employing bidirectional gated recurrent units in a domain adversarial learning paradigm. Likewise, Sun et al. [62] designed an architecture comprising parallel Transformer and 3D CNNs to obtain and combine temporal and spatial EEG features. More recently, Ma et al. [63] explored spatial-frequency shifted-window representations along with temporal self-attention for combining local and global EEG information.

TABLE 3 | Comparative overview of representative EEG preprocessing techniques relevant to emotion recognition.

Study	Approach	Preprocessing Method	Advantages	Limitations	Key Findings
Somers et al. [49]	Multichannel Wiener filtering with GEVD-based low-rank artifact covariance estimation	Multitype artifact suppression	Removes annotated artifact types while limiting distortion of neural EEG signals	Requires representative artifact annotations; performance and cost depend on filter settings	GEVD-based MWF outperformed evaluated ICA- and CCA-based methods on hybrid and real EEG
Ahmed et al. [58]	InvBase correction through inverse filtering of trial EEG spectra using resting-state baselines	Spectral baseline correction	Reduces baseline-related inter-individual variation before feature extraction	Requires baseline recordings; evaluated with fixed-size time windows	Improved valence and arousal classification over subtractive and no-baseline methods, including held-out-subject performance on DEAP
Sai et al. [59]	Wavelet–ICA with SVM-based identification of contaminated independent components	Automated hybrid artifact removal	Reduces reliance on manually selected thresholds for component identification	Requires pretrained SVM; validation focuses primarily on eye-blink artifacts	Achieved 99.1% artifact-component identification accuracy versus 95.2% for thresholding on the evaluated EEGLAB dataset
Issa and Juhasz [51]	Automatic ICA with selective wavelet cleaning of EOG-contaminated component intervals	Hybrid ICA–wavelet EOG removal	Preserves unaffected component intervals without dedicated reference EOG channels	Restricted to ocular artifacts; extensively contaminated components may be rejected	Outperformed evaluated component-rejection ICA and wICA methods in artifact-removal and signal-preservation metrics
Val-Calvo et al. [50]	EAWICA artifact removal with feature extraction, Savitzky–Golay smoothing, and offline chi-squared feature selection	Real-time EEG preprocessing pipeline	Supports low-latency artifact removal and emotion prediction using reduced EEG features	Lower artifact-removal fidelity than ICA-W in some synthetic comparisons; offline preparation is required	Reported 548.02 ms online prediction time and evaluated three-class emotion recognition on SEED

Apart from other methods, PSD and DE have proven popular in the analysis of specific frequency EEGs [64]. The power spectral density represents the signal power distribution among different frequencies whereas DE is the information theory-based measure of probability density of a continuous random signal. In the case of an EEG sample as Gaussian random variable, the expression of differential entropy is expressed as:

$$h(X) = \frac{1}{2} \ln \left(2\pi e \sigma_X^2\right), \tag{1}$$

where σ_X^2 denotes the variance of the EEG segment. Under the Gaussian assumption, DE is an affine function of logarithmic variance. For approximately zero-mean, band-pass-filtered EEG signals, variance is related to the signal power within the corresponding frequency band. Thus, DE provides a compact logarithmic representation of band-specific EEG activity.

Although PSD and DE are computationally manageable features, their effectiveness is influenced by signal segmentation and the selected analysis-window duration. Longer windows or temporal averaging may reduce sensitivity to transient changes, whereas shorter windows can increase the number of feature samples and affect spectral estimation. According to Ouyang et al. [64], the impact of the window length on the emotion recognition process depends on the feature extraction (PSD and DE features) from the SEED database. But overlapping windows do not always have to be used and depend on the type of experiment conducted.

Overall, EEG feature extraction has progressed from handcrafted signal descriptors toward learned multidimensional representations capable of modeling complementary spectral, spatial, and temporal information. Nevertheless, the effectiveness of these approaches depends on balancing discriminative information, feature redundancy, computational requirements, and sensitivity to recording variability. Below are discussed in detail different methods of extracting features temporally, spectrally, spatially, and in multiple domains.

2.8.4.1 | **Temporal Domain Features.** Temporal feature representations are indicative of the changes that occur in the EEG signal with respect to time, indicating the dynamic changes in neural responses based on emotions. Such features include statistical attributes and the dependency of the EEG signals at consecutive instances of time. Neural sequence learning models like recurrent and convolutional-recurrent models can effectively learn the temporal dependencies in emotion recognition tasks [18]. However, they require proper temporal segmentation and windowing of the EEG data.

2.8.4.2 | **Frequency Domain Features.** The Frequency Domain Features characterize the neural oscillation in the frequency bands of the EEG. Some of the examples of these kinds of features include Power Spectral Density (PSD), Band Power, and Differential Entropy (DE). These Frequency Domain Features act as the

efficient means to represent the Frequency Domain features of emotion processing [65, 66]. While these features can be generally applied in EEG-based emotion recognition, their effectiveness depends on preprocessing, spectral estimation parameters, and inter-subject variability.

2.8.4.3 | Time Frequency Domain Features. The time-frequency representations contain information about the changes in the frequency features of the EEG signal over time, which cannot be detected when averaging time-frequency analysis. There are methods such as the short time Fourier transform (STFT) and wavelet decompositions, which allow for local analysis of non-stationary EEG signals [66]. The efficiency of these methods depends on the choice of transformation parameters.

2.8.4.4 | Integration and Fusion Strategies Across Feature Domains. Time domain, frequency-domain, and spatial domain features help each other out in the aspects of dynamics, oscillations, and the dependency of EEG channels. Some of the widely used feature fusion techniques are concatenation, multi-dimensional feature representation, and neural network learning of the relationship between the feature domains [65, 67]. Feature fusion can potentially help in representing emotion-related EEG patterns, but this is dependent on various factors. It is difficult to have cross-subject generalization of the models.

Table 4 shows the comparison between traditional methods for extracting features from EEG data and deep learning based multidomain representation approaches. Traditional methods describe the temporal dynamics, spectral activities and time-frequency variations in terms of statistical features, higher order crossings, spectral analysis and signal decomposition [76, 77]. Recent methods consider the spatial, temporal and spectral information of the EEG signal through convolutional, recurrent and attention mechanisms [61]. Nevertheless, both types of methods face some trade-offs regarding the computational cost, discriminatory power and interpretability.

2.8.5 | Feature Selection

Feature selection aims to identify informative EEG features while reducing irrelevant or redundant information. EEG-based emotion recognition commonly involves high-dimensional feature representations obtained from multiple electrodes, frequency bands, and temporal segments. Although signal segmentation can generate numerous feature samples, the number of independent participants and recording trials may remain relatively limited. This imbalance can increase the risk of overfitting and make it difficult to identify emotion-discriminative patterns that generalize across recordings and individuals. Properly chosen feature subsets can lead to lower computational demands and enhanced classification accuracy and generalization ability, although this is contingent upon many factors [17].

Feature selection is particularly important for subject-independent EEG emotion recognition because the statistical distributions of EEG features may vary substantially across individuals. Features that discriminate emotional states within one subject may not remain equally informative for another. Consequently, subject-independent recognition requires representations that preserve emotional discriminability while remaining sufficiently consistent across participants. The problem was solved by Yin et al. [79] using locally robust feature selection with respect to similarity in individual distributions of features and emotion-discriminative data. The feature selection methods can be classified as filter, wrapper, or embedded according to the way features are evaluated and selected.

2.8.5.1 | Filter Methods. Dimensionality reduction using filter methods is done without taking into account any machine learning or deep learning models as this approach takes into consideration both statistical relevance and statistical redundancy of the variables independent of the learning model [80]. Popularly, these filters include information criterion based and correlation criterion based methods that can eliminate redundancy of variables during feature selection.

2.8.5.2 | Wrapper Methods. Evaluation of the performance of candidate subset of features is done using the classifier performance in wrapper methods. The methods include sequential forward selection (SFS), sequential backward selection (SBS), and evolutionary optimization. SFS has been noted to perform well in the classification of EEG emotions under certain circumstances [81]. This comes with an increase in computational complexity due to repetitive classifier evaluation.

2.8.5.3 | Embedded Methods. The Embedded approach combines feature selection in the learning phase, making it possible to learn feature importance from an optimization process. The examples of embedded approaches can be given as supervised orthogonal regression feature weighting and LASSO based selection. Unlike explicit feature selection, neural attention generally assigns adaptive importance weights without necessarily eliminating features. These approaches offer different trade-offs between computational efficiency, feature reduction, and model-dependent representation learning.

Table 5 shows the summary of the representative EEG feature selection methods and points out their methodology, advantages, and limitations. In filter based techniques, statistical measures like correlation and normalized mutual information are used to remove redundant features or channels, regardless of the classifier. On the other hand, wrapper-based approaches involve sequential feature selection and differential evolution, which assess the subsets in terms of their classification performance yet require additional computations. In embedded approaches, for instance, orthogonal regression and LASSO, feature selection is done as part of the model training, while in attention-based models, features and channels are weighted adaptively. From the review of the studies, it became evident that feature or channel selection could bring computational improvements and improve classification performance when applied under certain experimental conditions. Yet, the effectiveness of the techniques depends on particular datasets, selection criteria, classifiers, and evaluation strategies. Thus, feature selection should take into account the balance between dimensionality reduction, discriminative power, and complexity, as well as subject independent cross subject generalization should be confirmed by appropriate evaluation.

Fig. 6 illustrates the principal stages involved in acquiring and preparing EEG signals for emotion recognition. The process begins with experimental protocols in which participants are exposed to emotion eliciting stimuli, such as videos, audio recordings, and images, while their neural responses are captured

TABLE 4 | Comparative overview of representative EEG feature extraction and representation learning methods for emotion recognition.

Study	Approach	Domain Category	Advantages	Limitations	Key Findings
Li et al. [60]	3D feature representation using electrode maps, frequency bands, and dilated fully convolutional networks	Spatial–spectral; multiple feature types	Preserves electrode topology, inter-band relationships, and complementary feature information	Requires electrode-position mapping and construction of multifeature EEG arrays	Evaluated 3DFR-DFCN on DEAP and DREAMER under subject-dependent and subject-independent protocols
Lew et al. [61]	Regionally operated domain-adversarial network using Bi-GRU and attention	Spatial–temporal; domain adaptation	Learns temporal and interregional EEG relationships while reducing domain shift	Recognition results depend strongly on the subject-level separation and evaluation protocol	Demonstrated differences between subject-dependent, subject-independent, and subject-biased evaluations on DEAP and SEED-IV
Song et al. [68]	EEG Conformer combining temporal and spatial convolutions with Transformer self-attention	Spatial–temporal	Integrates local EEG representations with global temporal dependencies	Emotion experiments used SEED under a subject-dependent setting; subject-independent evaluation was absent	Demonstrated competitive multiclass emotion decoding on SEED and evaluated local–global feature learning
Jia et al. [69]	Two-stream spatial–spectral/temporal attention 3D dense network	Spatial–spectral–temporal	Learns discriminative patterns across brain regions, frequency bands, and temporal segments	Evaluated using subject-dependent sample splits on SEED and SEED-IV	Ablation experiments supported the contributions of two-stream fusion and attention mechanisms
Sun et al. [62]	Parallel Transformer and 3D-CNN with spatial and temporal feature fusion	Spatial–temporal	Extracts complementary information from two reconstructed EEG representations	Requires separate EEG representations and parallel processing branches	Reported improved classification performance against evaluated methods on DEAP, DREAMER, and SEED
Yudhana et al. [70]	FFT-based frequency features with k-nearest-neighbor classification	Spectral	Uses a straightforward frequency-domain representation with a single EEG channel	Evaluated on a single-channel recording protocol involving 50 participants	Reported 83.33% average accuracy for normal, focused, sad, and shocked conditions
Chen et al. [71]	Differential entropy extraction followed by LDA-based dimensionality reduction	Spectral; supervised projection	Combines band-specific EEG descriptors with discriminant feature reduction	Evaluated on a three-category emotion dataset rather than separate valence–arousal classification	Reported higher average classification accuracy than differential entropy alone, with reduced execution time
Lin et al. [72]	Spectral power and interhemispheric asymmetry feature analysis	Spatial–spectral	Captures emotion-related spectral differences across electrode pairs and frequency bands	Evaluated four music-induced emotion categories in a 26-participant study	Achieved 82.29% ± 3.06% average accuracy using spectral asymmetry features and SVM
Jenke et al. [73]	Comparative evaluation of temporal, spectral, and advanced EEG feature-extraction techniques	Multidomain	Systematically evaluates multiple EEG descriptors and feature-selection methods	High feature-to-sample ratio prevented statistical significance testing	HOC, HOS, and HHS features outperformed conventional spectral power features; multivariate selection performed slightly better than univariate selection
Murugappan et al. [74]	Discrete wavelet transform with energy, RMS, and recurring energy efficiency features	Time–frequency	Provides multiresolution EEG representations using wavelet-derived descriptors	Evaluated four emotion classes using recordings from only six participants	Reported effective four-class emotion discrimination using db4 wavelet features and fuzzy c-means clustering

Continued on next page

TABLE 4 | Comparative overview of representative EEG feature extraction and representation learning methods for emotion recognition (continued).

Study	Approach	Domain Category	Advantages	Limitations	Key Findings
Gupta et al. [75]	Flexible analytic wavelet transform with information-potential features and RF/SVM classifiers	Time–frequency	Extracts channel-specific information from flexible EEG sub-band decomposition	Experiments used selected EEG channels and specified decomposition parameters	Reported improved cross-subject, channel-specific emotion classification on SEED and DEAP relative to evaluated methods
Petrantonakis and Hadjileontiadis [76]	Higher-order crossing features with conventional classifiers	Temporal	Characterizes oscillatory EEG signal patterns for emotion discrimination	Evaluated 16 participants under controlled facial-expression emotion elicitation	Reported 83.33% six-class classification accuracy using combined-channel EEG and SVM
Wang et al. [77]	STFT-based electrode-frequency distribution maps with residual CNN and transfer learning	Spatial–frequency; time-windowed	Organizes frequency information across electrodes for convolutional feature learning	Cross-dataset adaptation was evaluated using available target-domain samples	Reported 90.59% on SEED and 82.84% in the DEAP transfer-learning experiment
Zheng and Lu [29]	Differential entropy features and DBN-based frequency-band and channel importance analysis	Spatial–spectral	Identifies emotion-informative frequency bands and reduced electrode configurations	Evaluation involved 15 participants recorded across two sessions	Reported 86.65% accuracy using 12 selected channels, compared with 83.99% using 62 channels
Yin et al. [78]	Differential entropy feature cubes with graph convolutional networks and LSTM fusion	Spatial–spectral–temporal	Combines graph-based inter-channel relationships with temporal feature modeling	Evaluated on DEAP using specified 6-s EEG segments	Reported subject-independent accuracies of 84.81% for valence and 85.27% for arousal

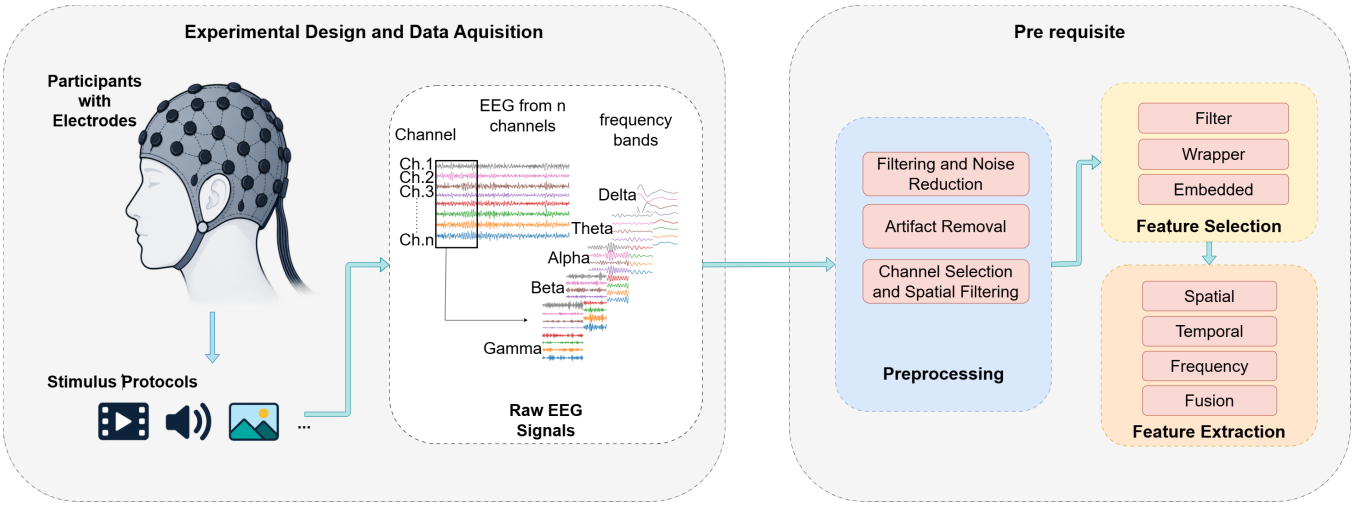


FIGURE 6 | Overview of the experimental design, EEG data acquisition, and signal processing stages for emotion recognition.

through multichannel EEG electrodes. The acquired signals contain time varying neural activity distributed across conventional frequency bands, including delta, theta, alpha, beta, and gamma. However, raw EEG recordings are susceptible to physiological artifacts, environmental interference, and irrelevant signal components, making appropriate preprocessing essential before extracting emotion related information. Common procedures include noise filtering, artifact removal, independent component analysis (ICA), wavelet based denoising, and spatial filtering. Following preprocessing, temporal, frequency domain, and spatial features can be extracted individually or combined through feature fusion to characterize complementary aspects of affective neural activity. Feature selection techniques, including filter, wrapper, and embedded approaches, can subsequently be employed to identify informative representations and reduce feature redundancy. These interconnected stages establish the foundation for reliable

TABLE 5 | Comparative overview of representative EEG feature selection techniques for emotion recognition.

Study	Approach	Selection Method	Advantages	Limitations	Key Findings
Xu et al. [82]	Supervised feature selection using orthogonal regression and feature weighting	Embedded	Reduces redundant information while retaining discriminative EEG features	Evaluated on 16 participants using random training/test partitions; subject-held-out generalization was not established	FSOR achieved 75.54% accuracy with 120 selected features, exceeding ReliefF and Information Gain in the reported comparison
Kannadasan et al. [83]	Differential-evolution-based feature selection with SVM classification	Wrapper	Optimizes candidate feature subsets using classifier-based fitness and redundancy reduction	Requires iterative optimization; evaluation was limited to DEAP	Reported 73.60% valence and 74.23% arousal accuracy in leave-one-participant-out evaluation, outperforming evaluated optimization baselines
Liu et al. [84]	EMD-based spectral centroid and LZC feature extraction followed by sequential backward selection	Wrapper	Captures multiscale EEG information while reducing redundant features	Evaluated on DEAP under specified frequency-band and temporal-window configurations	The complete EMD–SBS pipeline achieved 86.46% valence and 84.90% arousal accuracy using 1-s windows
Wang et al. [54]	NMI-based EEG channel selection using inter-channel connectivity	Filter-based channel selection	Substantially reduces the number of required EEG electrodes	Accuracy decreased slightly relative to all 32 channels; selection depends on NMI thresholds	Achieved 74.41% valence accuracy using 8 channels and 73.64% arousal accuracy using 10 channels, compared with 75.16% and 74.41% using 32 channels on DEAP
Chakladar and Chakraborty [80]	Correlation-based channel subset selection followed by higher-order statistical feature extraction	Filter-based channel selection	Reduces EEG input dimensionality from 13 to 4 channels	Evaluated using four emotional categories under the study's specified experimental protocol	Reported 82% four-class classification accuracy using four selected channels and an LDA classifier
Jenke et al. [73]	Comparative evaluation of EEG feature extraction and univariate/ multivariate selection methods	Filter-based comparison	Provides a systematic comparison of multiple EEG feature types and selection criteria	High feature-to-sample ratio prevented statistical significance assessment of the results	Multivariate selection performed slightly better than univariate selection in the evaluated dataset
Hatamikia et al. [81]	Sequential forward selection of Burg-method autoregressive EEG features using KNN, LDA, and QDA	Wrapper	Selects discriminative feature subsets based on classifier performance	Evaluated on DEAP using specified AR model orders and classifiers	SFS improved classification accuracy by approximately 10–15 percentage points over Davies–Bouldin-based selection in the reported comparisons
Kheirabadi and Omranpour [85]	EMD-derived statistical features with RBF-kernel transformation and LASSO-based selection	Embedded	Reduces feature dimensionality through sparsity-inducing regularization	Evaluation used 15 of 32 DEAP participants; results reflect the combined processing pipeline	Demonstrated LASSO-based feature selection within an EMD-based, cluster-assisted emotion classification framework

EEG based emotion classification and the development of computational models capable of learning discriminative emotional patterns.

2.9 | Challenges and Strategies in EEG Data Handling

The EEG based emotion recognition process can be hampered by high dimensional redundancy issues due to overfitting problems that arise due to the small amount of labeled training samples. However, data augmentation and artificial sample generation could address these problems, as long as the integrity of the evaluation process is maintained.

2.9.1 | Feature Redundancy

EEG features may contain redundant information due to inter-channel dependencies, particularly among neighboring electrodes. Filter, wrapper, and embedded methods provide different mechanisms for reducing dimensionality. Orthogonal-regression-based feature selection has been investigated to reduce redundant information while retaining discriminative EEG features [82]. However, feature reduction does not guarantee improved generalization. Furthermore, evaluations that distribute highly correlated segments from the same trial across training and testing sets can produce overly optimistic results [61]. Therefore, feature selection should be fitted exclusively on training data, with independent evaluation partitions preserved.

2.9.2 | Limited Data Challenges

Collecting sufficiently large and diverse emotional EEG datasets is challenging because of recording costs, participant recruitment, and the difficulty of obtaining labeled affective responses [86]. The two techniques provide a complementary way of enhancing the training sample size.

2.9.2.1 | Augmenting Data Samples. Data Augmentation is the process of generating more training instances from transformations of existing EEG recordings. Transformations need to maintain emotion-related features and should not conflict with the corresponding labelings. Some research on masking and reconstructing EEG representations has been performed while modeling both spatial and temporal structure of EEGs [87]. To avoid any information leaks, augmentation needs to be limited to training splits only.

2.9.2.2 | Generating Synthetic Data. Generative models learn the properties of the training distribution for EEGs to generate more samples and features. CWGAN has been used to synthesize class-conditioned differential entropy features where the generated samples were tested before being added to the training set [86]. Likewise, GANSER has combined adversarial training with masked signal reconstruction and self supervised learning to synthesize EEG data [87]. However, the quality of the synthetic data and any improvement in classification performance needs to be tested in an experiment-specific setting rather than generalized.

In summary, successful management of the EEG data involves finding the right balance between feature reduction and data augmentation while maintaining the emotion related information.

3 | Modeling and Evaluation Frameworks

Emotion recognition from EEG entails converting preprocessed neural signals to representations of emotions and classification outputs. Some deep learning methods may learn spatial, spectral, and temporal information using convolutional, recurrent, graph, and attention mechanisms [88, 89]. Nevertheless, the efficacy of these methods is not only dependent on the architecture but also on the experimental setup that includes trial-, session-, and subject-level splits between training and testing sets.

3.1 | Subject-Dependent Analysis

Subject-dependent analysis is concerned with emotion detection within a particular subject, with data used to train and test the model coming from the same subject. Models may be trained separately for each participant or adapted using subject-specific recordings. This setting allows the learning of individual EEG characteristics but requires careful validation to distinguish emotion-related patterns from trial-specific dependencies. In particular, randomly distributing temporally correlated segments from the same trial across training and testing sets may produce optimistic performance estimates. Trial-wise cross-validation and leave-one-trial-out evaluation address this concern by keeping entire trials separate during evaluation [88].

Deep learning architectures have expanded subject-dependent emotion recognition by integrating complementary EEG representations. While graph-based methods identify interactions between electrodes, convolution and recurrence learn spatial and temporal relationships, respectively. For example, the SOG-PCN architecture combines frequency-specific channel graphs, pseudo-3D convolution, attention, and LSTM-based temporal modeling [89]. Its evaluation on SEED includes both randomized-fold and leave-one-subject-out protocols. However, the reported results should be interpreted within their respective data-partitioning procedures rather than regarded as evidence of unrestricted cross-subject generalization.

Temporal learning has also been investigated through residual recurrent architectures. Multimodal residual LSTM models incorporate weight sharing to learn temporal relationships between EEG and other physiological signals [90]. Although multimodal integration can exploit complementary physiological information, these systems differ from EEG-only approaches in their data requirements and should not be compared without accounting for the additional sensing modalities.

Hybrid convolutional and Transformer architectures provide another strategy for learning spatial and temporal EEG representations. Interpretable models like ERTNet combine temporal and spatial convolutions with Transformer encoding to extract discriminative features, and then examine their impact on the task of emotion recognition [91]. Nevertheless, the experiments originally conducted show significant differences in performance when compared using subject-dependent and leave-one-subject-out approaches. This indicates that interpretable feature representations do not necessarily provide equivalent recognition performance on unseen participants.

A regularized subject-dependent learning objective can be expressed as

$$\mathcal{L}_{\mathrm{SD}}^{(u)} = \frac{1}{|\mathcal{D}_{\mathrm{tr}}^{(u)}|} \sum_{(x,y)\in\mathcal{D}_{\mathrm{tr}}^{(u)}} \ell\left(f_{\theta^{(u)}}(x), y\right) + \gamma\Omega\left(\theta^{(u)}\right), \tag{2}$$

where $\mathcal{D}_{\mathrm{tr}}^{(u)}$ denotes the training data for participant u, $f_{\theta^{(u)}}$ represents a subject-specific or subject-adapted classifier, and $\ell(\cdot)$ is the classification loss. The regularization term $\Omega(\theta^{(u)})$, such as the squared L_2 norm, penalizes model complexity, while γ controls its contribution to the objective. This equation is a general model of regularized learning with respect to subject-dependent data instead of being a goal associated with any reviewed structure.

Graph-based architectures have been investigated to capture spatial dependencies and functional relationships among EEG channels. Dynamical graph convolutional neural networks (DGCNNs) learn inter-channel relationships through an adaptive adjacency matrix, achieving 90.40% subject-dependent and 79.95% subject-independent classification accuracy on SEED under the reported evaluation protocols [92]. Regularized graph neural networks (RGNNs) incorporate neuroscience-informed channel connectivity, node-wise domain adversarial training, and emotion-aware distribution learning to address inter-subject variability and noisy emotion labels [9]. Evaluations on SEED and SEED-IV demonstrate the potential of graph-based representations for emotion recognition under both subject-dependent and subject-independent settings. Nevertheless, their effectiveness must be interpreted according to the datasets and evaluation protocols employed. Overall, subject-dependent analysis proves to be a valuable tool for analyzing personalized EEG emotion recognition and neural response pattern of the individual. On the other hand, high precision for a certain individual in terms of classification does not ensure generalization over subjects and over recording sessions.

3.2 | Subject Independent Analysis

Subject independent EEG emotion recognition seeks to decode affective states from participants whose emotion labels were not used to train the original classifier. Its principal challenge is inter individual variability meaning EEG recordings from different participants may exhibit different feature distributions even when elicited by similar emotional stimuli. Session related nonstationarity and changes in acquisition conditions introduce further discrepancies, so performance measured within a participant cannot be assumed to transfer to a new participant [93, 94]. Transfer learning consequently seeks representations and decision rules that retain emotion discriminative information while reducing sensitivity to differences between source and target recordings.

This setting is important for practical affective brain computer interfaces because collecting a complete labeled calibration dataset for every incoming user is burdensome. Nevertheless, subject independent describes the separation of emotion labels across participants, not necessarily the absence of all target domain observations during model development. In unsupervised domain adaptation, unlabeled EEG from the target participant is accessible for alignment, approaches with labeled target calibration instead operate in a few shot or supervised adaptation setting, and domain generalization assumes that no target recordings are used to learn the deployed model [95, 96]. Accordingly, these protocols must be distinguished when interpreting reported performance.

Initial deep adaptation approaches focused on the discrepancy between source and target feature distributions. Li et al. [93] investigated a deep adaptation network (DAN) that combines supervised emotion classification with multi kernel maximum mean discrepancy (MK-MMD) losses at intermediate network layers. Evaluations on SEED and SEED-IV showed improvements over the comparison methods under the authors' cross subject protocol. A complementary approach was developed by Li et al. [97], who combined adversarial matching of marginal representations in earlier layers with association reinforcement for class conditional alignment in later layers. Their experiments on SEED and DEAP addressed cross subject and cross session transfer. Together, these studies distinguish global feature alignment from approaches that also consider the structure of emotion classes.

Source domain heterogeneity introduces a further difficulty: recordings from different individuals may not be equally useful for a particular target participant. Li et al. [98] proposed multi source transfer using source selection, style transfer mapping, and integration of source classifiers. Source selection relies on a small labeled target calibration set, the authors also explored supervised and semi supervised variants of the mapping stage. Thus, the method reduces the labeled data requirement but is not calibration free. More recently, Yang et al. [99] proposed spectral spatial attention alignment for multi source domain adaptation (S^2A^2-MSDA). Their framework aligns attention related representations across EEG frequency bands and electrode locations, combines them with source target feature alignment, and exploits target pseudo labels. It was evaluated for cross session recognition on SEED and cross subject recognition on SEED-IV. These two studies demonstrate distinct ways of handling multiple sources, rather than establishing a single universally optimal fusion strategy.

Models that preserve relational or class specific information offer alternatives to aligning only overall feature distributions. Xu et al. [100] developed the domain adversarial graph attention model (DAGAM), representing EEG channels through a graph informed by electrode topology, extracting salient features by self attention pooling, and aligning pooled representations adversarially. Under the reported leave one subject out evaluations, DAGAM achieved 92.59% on SEED and 80.74% on SEED-IV. Its authors also reported lower computational complexity than the compared node level adversarial formulation; a general claim that this architecture necessarily incurs unusually high cost would therefore be unsupported. Zhou et al. [101] proposed prototypical representation based pairwise learning (PR-PL), jointly using emotion category prototypes, domain adversarial learning, supervised source pairs, and confidence selected pseudo labeled target pairs. Their SEED and SEED-IV experiments addressed both inter subject variation and imperfect emotion annotations.

Other work investigates the relationships between EEG samples without relying exclusively on a domain discriminator. Shen et al. [26] introduced contrastive learning for inter subject alignment (CLISA): EEG responses recorded from different individuals during the same stimulus segment form positive contrastive pairs, whereas responses to different stimuli form negative pairs. The learned representations were evaluated on THU-EP and SEED. Importantly, this stimulus based pairing is not equivalent to treating every pair of recordings with the same emotion label as a positive pair. Peng et al. [102] addressed target label estimation through joint feature adaptation and graph adaptive label propagation (JAGP), integrating distribution alignment, graph learning, and emotion prediction within a common optimization framework. Their evaluations used SEED-IV and SEED-V. These approaches bring out the importance of maintaining the semantic or relational structure during the process of transfer.

Cross dataset experiments reveal challenges that are not evident in within dataset experiments. Rayatdoost and Soleymani [25] studied the transferability of EEG emotion features across DEAP, MAHNOB-HCI, and DAI-EF, stating difficult cross dataset scenarios and studying the possible contamination by facial muscle activity. Cross dataset analysis using the same architecture,

that is, pretrained convolutional networks for windows and normalization of EEG signals, was performed by Cimtay and Ekmekcioglu [103], achieving an average accuracy of 58.10% while evaluating a network trained on SEED with DEAP. Imtiaz and Khan [104] developed an unsupervised adaptation framework combining gradual proximity guided target data selection (GPTDS) with prediction confidence aware test time augmentation (PC-TTA); they reported 67.44% for DEAP to SEED and 59.68% for SEED to DEAP transfer. These percentages describe different model and evaluation protocols and should not be interpreted as directly interchangeable estimates of cross dataset robustness.

Overall, subject independent emotion recognition has advanced through discrepancy minimization, adversarial learning, multi-source alignment, graph based representation, pairwise learning, and contrastive training. However, reported performance remains conditional on the target information available during adaptation, the emotion label scheme, the electrode representation, and the partitioning of recordings. Claims of generalization must therefore be tied to subject held out, session held out, or cross dataset evaluations that match the intended deployment scenario.

3.2.1 | Transfer Learning Framework

Transfer learning applies information acquired from existing EEG recordings to a related subject, session, device, or task. Reviews of EEG based brain computer interfaces identify individual variability, within subject nonstationarity, and limited calibration data as major motivations for transferring learned representations or model parameters [94, 105]. Three settings are particularly relevant to EEG emotion recognition: domain adaptation uses available target observations to reduce distribution mismatch; domain generalization seeks transfer to unseen target domains without target data during training; and fine tuning updates a pretrained model using target specific calibration. Semi supervised learning, which combines labeled and unlabeled observations, is related but should not automatically be classified as parameter efficient domain adaptation.

3.2.1.1 | Domain Adaptation. Domain adaptation (DA) seeks to improve a classifier's performance when the source and target observations have different distributions but support a related prediction task. For EEG emotion recognition, domains may represent different participants, sessions, or independently collected datasets. Distribution mismatch can involve the marginal EEG feature distribution $P(X)$, class proportions $P(Y)$, or the conditional relationship $P(Y \mid X)$, and aligning marginal distributions alone cannot guarantee matching emotion labels [97, 105]. For a conventional unsupervised DA problem with labeled source observations and unlabeled target observations, let

$$\mathcal{D}_s = \{(x_i^s, y_i^s)\}_{i=1}^{n_s}, \qquad \mathcal{D}_t = \{x_j^t\}_{j=1}^{n_t}. \tag{3}$$

When the domains have a common feature and emotion label space, their joint distributions may differ:

$$P_s(X, Y) \neq P_t(X, Y). \tag{4}$$

3.2.1.1.1 | Discrepancy Based Methods. Discrepancy based approaches explicitly compare statistics of source and target representations. Li et al. [93] applied MK-MMD at multiple network layers for cross subject EEG emotion recognition. The original Deep CORAL method of Sun and Saenko [106], developed outside the EEG domain, instead minimizes differences between source and target activation covariance matrices. It provides a general methodological foundation for covariance alignment, not an EEG specific experimental result. In affective EEG, Li et al. [97] extended the alignment perspective by considering both marginal and conditional distribution similarity, using adversarial learning and association reinforcement in different layers.

Not all EEG adaptation systems fit exclusively within a single discrepancy based category. Bao et al. [107] introduced a two level domain adaptation neural network (TDANN), extracting topological EEG features and combining MMD based distribution reduction with domain adversarial learning that encourages class centred representations. The study evaluated cross day and cross subject transfer using a self collected dataset and SEED, reporting 74.93% cross day and 87.9% cross subject accuracy on SEED under their respective protocols. TDANN is therefore more accurately described as a discrepancy adversarial hybrid than as a purely MMD based method.

Such methods are appealing because their alignment criteria are explicitly defined. Nevertheless, minimizing a discrepancy in feature space is not sufficient evidence that target emotion classes remain separable. Comparisons should consider whether an approach aligns only marginal statistics, incorporates conditional information, or uses additional domain specific supervision.

A general discrepancy based objective can be written as

$$\begin{aligned} \min_{\theta_f, \theta_c} \; \mathcal{L}_{\mathrm{DA}} &= \mathcal{L}_{\mathrm{cls}}(\mathcal{D}_s; \theta_f, \theta_c) \\ &\quad + \lambda \, \mathcal{L}_{\mathrm{align}}(\mathcal{D}_s, \mathcal{D}_t; \theta_f), \end{aligned} \tag{5}$$

where θ_f and θ_c parameterize the feature extractor and emotion classifier, respectively; $\mathcal{L}_{\mathrm{cls}}$ measures source label prediction error; and $\mathcal{L}_{\mathrm{align}}$ penalizes a chosen discrepancy between source and target representations. The coefficient λ balances classification against alignment. This expression is a generic synthesis of discrepancy based DA and should not be treated as the exact loss used by every reviewed method.

3.2.1.1.2 | Adversarial Based Methods. Adversarial DA combines a feature extractor with a domain discriminator that tries to identify the origin of learned representations. Ganin et al. [108] established the domain adversarial neural network (DANN) framework, using labeled source data, unlabeled target data, and gradient reversal. Their original experiments concerned image classification, text sentiment, and person re identification, rather than EEG. Jin et al. [109] subsequently applied a gradient reversal based domain adaptation network to EEG emotion recognition, reporting 79.19% mean accuracy in their evaluated cross subject setting.

A standard adversarial objective, with domain loss defined as the discriminator's classification error, is

$$\min_{\theta_f, \theta_c} \max_{\theta_d} \left[\mathcal{L}_{\mathrm{cls}}(\theta_f, \theta_c) - \lambda \mathcal{L}_{\mathrm{dom}}(\theta_f, \theta_d) \right], \tag{6}$$

where θ_d represents the discriminator parameters. For features $h_i^s = F_{\theta_f}(x_i^s)$ and $h_j^t = F_{\theta_f}(x_j^t)$, one possible domain loss is

$$\mathcal{L}_{\text{dom}} = \frac{1}{n_s}\sum_{i=1}^{n_s} \ell_d\big(D_{\theta_d}(h_i^s), 0\big) + \frac{1}{n_t}\sum_{j=1}^{n_t} \ell_d\Big(D_{\theta_d}(h_j^t), 1\Big), \tag{7}$$

where ℓ_d denotes a domain classification loss, commonly binary cross entropy. During training, the discriminator is encouraged to classify source and target representations correctly, while the feature extractor receives an opposing gradient and is simultaneously optimized to retain source label discrimination [108].

EEG specific adversarial methods instantiate this idea in different ways. Luo et al. [110] introduced Wasserstein GAN domain adaptation (WGANDA), with separate source and target feature mappings and a gradient penalized adversarial objective. Their leave one subject out experiments reported 87.07% on SEED, together with 66.85% arousal and 67.99% valence accuracy on DEAP. The gradient penalty was introduced to improve optimization behaviour; it should not be interpreted as proof that adversarial instability is eliminated in every setting.

Zhao et al. [95] developed plug and play adaptation through shared emotional and private subject specific encoders, combining a shared classifier with source specific classifiers weighted by similarity to the incoming participant. Calibration uses a small amount of unlabeled target EEG, with the authors reporting approximately one minute calibration on SEED. In contrast, Yang et al. [99] addressed multisource cross session and cross subject transfer through alignment of spectral spatial attention patterns, feature distributions, and target pseudo label information. Their method cannot be reduced to domain confusion alone because it explicitly incorporates attention consistency and conditional semantic information.

Graph structured adversarial alignment has also been investigated. DAGAM combines electrode topology graphs, self attention pooling, and a discriminator operating on pooled graph features [100]. Its original experiments reported improved recognition on SEED and SEED-IV and a computational advantage relative to the comparator discussed by its authors. Consequently, limitations regarding inconsistent channel layouts or other recording montages should be framed as untested transfer conditions rather than established failures of DAGAM.

Taken together, adversarial EEG adaptation methods differ in their feature encoders, alignment levels, target calibration requirements, and additional objectives. Their outcomes should be interpreted within these conditions, rather than assuming that adversarial learning always dominates discrepancy based methods.

3.2.1.1.3 | Prototype, Graph, and Similarity Based Methods. Class aware approaches aim to maintain emotion specific structure while mitigating distribution mismatch. Zhou et al. [101] developed PR-PL, which uses source derived emotion prototypes and supervised source pairs together with pseudo labeled target pairs selected by adaptive similarity thresholds. The method combines this pairwise learning with adversarial alignment and was evaluated on SEED and SEED-IV under four cross validation protocols. Peng et al. [102] proposed JAGP, jointly optimizing feature adaptation, a graph that is updated during learning, and propagation of estimated emotion labels for unlabeled target observations. The authors investigated the method on SEED-IV and SEED-V and also analyzed the importance of the learned channel and frequency band representations.

Contrastive alignment offers a related but distinct strategy. CLISA learns spatiotemporal EEG representations by increasing similarity between recordings obtained from different participants during the same stimulus segment and decreasing similarity for different stimulus segments [26]. Its positive pairs arise from stimulus correspondence, not simply from matching emotion labels. These models address complementary aspects of transfer: prototypes characterize emotion classes, graph propagation exploits relationships among observations, and contrastive learning exploits structured correspondence between recordings. The quality of pseudo labels or pair assignments must be considered when assessing their applicability to a new experimental setting.

3.2.1.1.4 | Source Free Domain Adaptation. Conventional unsupervised DA generally requires source recordings to remain accessible alongside unlabeled target observations while the adaptation model is trained. Source free DA instead adapts a previously trained source model using target data without retaining the original source samples. This distinction is relevant when data sharing restrictions, privacy concerns, or storage constraints prevent source EEG data from being used at deployment.

Imtiaz and Khan [111] investigated this setting in an available manuscript describing dual loss adaptive regularization (DLAR) and localized consistency learning (LCL). Their framework uses confidence dependent prediction regularization and relationships among reliable target neighbours to limit unreliable pseudo label updates; it also employs confidence aware test time augmentation. For the evaluated DEAP to SEED and SEED to DEAP transfers, the manuscript reported 65.84% and 58.99% accuracy, respectively. These are results from the uploaded manuscript and should not be presented as independently verified final journal findings.

By contrast, the earlier GPTDS and PC-TTA framework of Imtiaz and Khan [104] is an unsupervised cross-dataset adaptation method that uses source domain information during adaptation. Although both studies address unlabeled target observations and substantial dataset differences, only the former explicitly investigates source free adaptation. This difference should be preserved in comparisons of deployment requirements.

3.2.1.2 | Domain Generalization. Domain generalization (DG) seeks to develop models that perform well on previously unseen domains without access to the target domain recordings during model development. Zhou et al. [96] review strategies including invariant representation learning, augmentation, meta learning, and ensemble learning. Because that survey covers general machine learning rather than an EEG specific benchmark, it supports the taxonomy but not a quantitative claim about EEG emotion recognition.

For affective EEG, a domain generalization claim requires an evaluation in which recordings from the target participant, session, or dataset are withheld from all training and adaptation procedures. This is a stricter information constraint than unlabeled target DA or brief target calibration. For example, the contrastive inter subject representations examined by Shen et al. [26] were tested for transfer to unseen participants and stimuli; their experiments illustrate one representation learning route toward

greater transferability. However, no architecture should be described as inherently domain general simply because it achieves high within dataset accuracy. Explicitly reporting the use of target data and the held out unit is indispensable.

3.2.1.3 | Fine Tuning Strategies. Fine tuning updates a pretrained network using a limited amount of target domain calibration data. In supervised EEG emotion recognition, these data have emotion labels and are distinct from the target examples reserved for final testing. Depending on the protocol, only selected layers or a larger portion of the pretrained parameters may be updated [94, 105]. A generic parameter regularized objective is

$$\min_{\theta} \mathcal{L}_{\mathrm{FT}} = \frac{1}{|\mathcal{D}_t^{\ell}|} \sum_{(x,y)\in\mathcal{D}_t^{\ell}} \ell(f_\theta(x), y) + \mu\|\theta - \theta_s\|_2^2, \tag{8}$$

where $\mathcal{D}_t^{\ell}$ denotes the labeled target calibration subset, θ_s the pretrained parameters, and μ the weight assigned to preserving proximity to those parameters. This is an illustrative regularized fine tuning formulation, not the exact loss function of every study discussed below.

Lu et al. [112] developed a hybrid domain adaptation and few shot fine tuning network (DFF-Net). Their Emo-DA module first adapts a Vision Transformer based feature extractor using source and target data, after which labeled target observations are used for fine tuning. The authors reported mean accuracies of 93.37% on SEED and 82.32% on SEED-IV. Their ablation study found that the complete framework outperformed either DA only or fine tuning only variants; specifically, fine tuning alone achieved 88.09% and 78.31% on the respective datasets under the study's evaluation conditions. Since target labels contribute to fine tuning, these figures must not be compared directly with calibration free subject independent results.

Transfer across recording collections has also been explored using fine tuning. Rayatdoost and Soleymani [25] examined convolutional transfer from their DAI-EF corpus to MAHNOB-HCI, updating a pretrained network on the target corpus. Such experiments illustrate how target specific adaptation may improve cross corpus recognition when appropriate data are available, although the resulting improvement is conditional on the selected datasets and training setup. Source selection and style transfer strategies using a small labeled target calibration set, as in Li et al. [98], offer a related route to exploiting previously learned subject information without training a new model entirely from scratch.

Overall, robust cross subject and cross dataset EEG emotion recognition requires more than maximizing benchmark accuracy. Discrepancy minimization, adversarial adaptation, class aware representation learning, source free adaptation, domain generalization, and supervised fine tuning rely on different information assumptions. Accordingly, claims of transferability should specify whether target data or labels were used, whether source recordings were available during adaptation, and whether evaluation excluded entire subjects, sessions, or datasets. Such protocol transparency is essential for judging the reproducibility and practical relevance of reported improvements.

3.3 | Machine Learning and Deep Learning Methods

Machine learning and deep learning provide complementary approaches to mapping EEG recordings onto discrete emotion categories or dimensional affective responses. Conventional ML pipelines commonly separate signal preparation, feature extraction, and classifier training, whereas DL can learn task relevant representations through successive nonlinear transformations of raw or preprocessed EEG. This distinction is not absolute: deep networks may also operate on engineered spectral features, and their use does not eliminate the need for preprocessing or careful validation [22, 23]. Reviews of physiological signal based emotion recognition additionally emphasize the dependence of recognition methods on signal properties, feature representation, and classifier design [113].

Model selection matters because EEG is nonstationary, susceptible to physiological and environmental artifacts, and variable across individuals. Greater representation capacity may help capture complex patterns, but it does not by itself establish improved performance on new participants or recording sessions [23, 94]. Consequently, comparisons between shallow and deep approaches must consider the input representation, supervision, preprocessing, and evaluation protocol, rather than architecture alone.

An EEG window can be denoted by $\mathbf{X}_i \in \mathbb{R}^{C\times T}$, where C is the number of recorded channels and T the number of time samples. For discrete emotion classification, a possible supervised training objective is

$$\min_{\theta} \frac{1}{N}\sum_{i=1}^{N} \mathcal{L}_{\mathrm{CE}}(f_\theta(\mathbf{X}_i), y_i) + \lambda\|\theta\|_2^2, \tag{9}$$

where f_θ produces predicted class probabilities, y_i is an emotion label, N is the training sample count, and λ controls an optional L_2 penalty. Equation (9) is an illustrative regularized classifier objective, not the exact loss used in every reviewed study. Continuous valence or arousal prediction instead requires a loss appropriate to regression, while other classification frameworks can add complementary representation learning terms. For example, Li et al. [7] combined cross entropy and supervised contrastive losses in an efficient convolutional network, explicitly encouraging same class EEG representations to become closer and different class representations to separate.

The principal architectural distinction is how representations are constructed and constrained. CNN based encoders learn patterns through shared convolutional filters; Li et al. [7] replaced conventional convolutions with a more efficient block and evaluated the resulting ECNN-C framework on DEAP and DREAMER. Graph networks instead represent electrode relationships explicitly. Zhong et al. [9] proposed a regularized graph neural network with biologically informed channel connections, node wise domain adversarial training, and emotion aware distribution learning, evaluating both subject dependent and subject independent emotion recognition on SEED and SEED-IV. These are different inductive biases rather than evidence that one model family is universally superior.

Evaluation design is equally consequential. Multisource transfer studies demonstrate that models may require different amounts of labeled target calibration when applied to new individuals [98].

More generally, transfer learning reviews distinguish cross subject, cross session, and cross device changes in EEG distributions [94]. Lew et al. [61] explicitly compared subject dependent, subject independent, and subject biased protocols, showing why random assignment of correlated segments from the same trial can give misleadingly favorable results. Accordingly, splitting should be performed at the unit appropriate to the intended claim, and preprocessing parameters, feature selection, and hyperparameter tuning should be learned without access to the final test partition. Accuracy should be interpreted together with the partitioning procedure and the variability of outcomes across participants, rather than treated as a self sufficient measure of generalization. Taken together, the reviewed evidence supports treating feature engineering and end to end representation learning as complementary design choices. Their relative value must be established under comparable data preparation and evaluation conditions; an expressive architecture alone cannot resolve artifacts, limited supervision, or inter individual distribution shift.

3.3.1 | Shallow Models

Shallow EEG emotion recognition models generally use an explicit feature map ϕ to transform a recording window into a vector $\mathbf{z} = \phi(\mathbf{X})$, followed by a classifier such as a support vector machine, k nearest neighbor classifier, or another conventional learner. The feature map can summarize signal amplitudes, oscillatory activity, time frequency characteristics, or relationships among electrodes [22]. Because these quantities are defined before classifier training, their contribution can be inspected separately; however, explicit feature definitions do not automatically guarantee that an observed association is emotion specific or robust across recording settings [23].

Time frequency and nonlinear descriptors illustrate the flexibility of conventional pipelines. Liu et al. [84] used empirical mode decomposition (EMD) to obtain multiscale oscillatory components, extracted spectral centroid and Lempel Ziv complexity descriptors, and then applied sequential backward selection (SBS) before classification. The EMD representation can be expressed as

$$x_c(t) = \sum_{m=1}^{M_c} \mathrm{IMF}_{c,m}(t) + r_c(t), \tag{10}$$

where $\mathrm{IMF}_{c,m}$ denotes an intrinsic mode function of channel c and r_c the remaining component. In the authors' DEAP experiments, a 1 s analysis window with k nearest neighbor classification yielded 86.46% valence accuracy and 84.90% arousal accuracy. These are results for the complete EMD feature selection classifier pipeline under the study's protocol; they do not isolate the contribution of EMD or SBS individually.

Discrete wavelet analysis provides a distinct time frequency alternative to EMD. Murugappan et al. [74] extracted wavelet derived statistics from multichannel EEG and used fuzzy c means clustering to examine four induced emotional states (happiness, disgust, surprise, and fear). Their study compared 63 and 24 channel configurations using recordings from six participants. It therefore demonstrates a specific wavelet based processing pipeline but cannot, on its own, establish generalization to larger or unseen populations.

A different family of compact features describes the distribution of band limited EEG activity. Under a Gaussian approximation with band specific variance σ_b^2, differential entropy is

$$h_b = \frac{1}{2}\ln\left(2\pi e\,\sigma_b^2\right). \tag{11}$$

Chen et al. [71] combined differential entropy extraction with linear discriminant analysis for dimensionality reduction and evaluated the resulting features using several conventional classifiers on a three class EEG emotion dataset. Their reported gains concern the combined DE-LDA processing procedure rather than differential entropy in isolation. This distinction matters because a feature representation and the method used to select or project it can jointly determine classification outcomes.

Handcrafted representations need not be restricted to independent electrode wise statistics. Li et al. [114] constructed functional connectivity networks using phase locking values and combined network measures with local power related activation features. Their feature selection procedure ranks features using an F score and evaluates candidate subsets with classifier dependent selection, including support vector machines and graph regularized extreme learning machines. Experiments on three emotional EEG databases favored the fused activation connectivity representations over the individual feature families under the reported conditions. This paper does not establish that its selected features remain stable under arbitrary session drift, and its selection strategy should not be described as L_1 regularized regression.

Sparsity based feature selection is a separate option. For example, a generic least squares selection objective with an L_1 penalty is

$$\min_{\mathbf{w}} \|\mathbf{y} - \mathbf{Z}\mathbf{w}\|_2^2 + \alpha\|\mathbf{w}\|_1, \tag{12}$$

where $\mathbf{Z}$ contains extracted training features, $\mathbf{y}$ contains encoded targets, and α controls sparsity. This is an illustrative LASSO type formulation, not the feature selection objective used by Li et al. Kheirabadi and Omranpour [85] employed LASSO within a larger EMD derived, kernel transformed, cluster assisted EEG emotion classification pipeline; consequently, its reported classification outcomes cannot be attributed to the sparse penalty alone.

Other conventional classifiers have been adapted to the structure of EEG emotion datasets. Wankhade and Doye [115] described an improved k nearest neighbor predictor that varies the neighbor count by class, within a pipeline comprising preprocessing, feature extraction, and selection. Their evaluation used SEED based experiments and compared the proposed predictor with other classical classifiers. The work illustrates a classifier design alternative; it does not provide an experimental demonstration that the method is intrinsically more interpretable or better calibrated on unseen datasets.

Signal quality is a separate determinant of the reliability of engineered features. Lai et al. [116] reviewed contamination from eye movements, muscle activity, and other noise sources, noting that direct removal of contaminated components can also discard useful EEG information. Sai et al. [59] combined independent component analysis, an SVM for identifying artifact containing components, and wavelet based cleaning, with experiments focused primarily on eye blink contamination. Phadikar et al. [52] likewise addressed eyeblink artifacts, but used an SVM to identify affected recordings followed by discrete wavelet decomposition and metaheuristically optimized thresholds for reconstruction.

Neither study proposed EEG emotion feature or electrode selection, and their artifact removal findings should not be misrepresented as improvements in emotion classification accuracy.

Quality assessment can also precede artifact correction. Witteveen et al. [117] compared pragmatic threshold based and regression based indicators for identifying contaminated EEG segments in wearable recordings. Their evaluation included uncontrolled laboratory and free living measurements and reported over 83% average accuracy for artifact detection by the proposed approaches. These results concern EEG signal quality estimation, not the calibration or confidence reliability of an emotion classifier. At a broader application level, Egger et al. [118] reviewed physiological emotion recognition modalities and emphasized that the suitability of EEG and alternatives depends on acquisition conditions, motion sensitivity, and the intended application; the review does not establish shallow classifiers as universally preferable to deep models.

Overall, shallow pipelines provide explicit choices of features, feature selection procedures, and classifiers that can be examined individually. Their strengths and limitations are conditional on the recording quality, available labels, evaluation setting, and the relationship between features and affective states. In particular, the evidence does not justify a universal ranking of shallow versus deep models for small datasets, interpretability, or computational efficiency. Comparisons should instead report the complete processing pipeline and test whether its apparent advantages persist under appropriately separated trials, participants, and recording conditions [23, 22].

3.3.2 | Deep Learning Models

Deep learning provides a means of learning representations and emotion classifiers from EEG observations within a shared computational framework. Depending on the architecture, the input may consist of sampled EEG signals, spectral features, electrode maps, or other representations derived from the recordings. Convolutional networks, recurrent networks, and graph networks emphasize different structures in these inputs, while attention and representation learning objectives provide additional mechanisms for processing informative patterns [23, 9]. These capabilities should not be interpreted as an automatic advantage over conventional feature engineering: the systematic review by Roy et al. examined deep EEG methods across several application areas and noted substantial variation in preprocessing, training data, and reproducibility [23].

The relevance of deep models to EEG emotion recognition follows from the fact that affective information may be distributed across electrodes, frequency components, and successive time intervals. Their performance nevertheless depends on the recording and training conditions. Reviews of EEG emotion recognition document variation in emotional stimuli, electrode configurations, feature representations, classifiers, and evaluation procedures [35]. Moreover, artifacts originating from eye or muscle activity can contaminate the measured signals, so learned representations should not automatically be interpreted as exclusively neural or emotion related [116].

Architectural choices involve identifiable methodological differences. Li et al. [7] developed an efficient convolutional network with supervised contrastive learning (ECNN-C), using an economical convolutional block and a combination of classification and contrastive objectives. Zhong et al. [9] instead modeled relationships between EEG channels through a regularized graph neural network (RGNN), incorporating a topology informed adjacency matrix and regularizers designed for subject variation and uncertain emotion labels. These experiments establish the feasibility of distinct representation learning strategies on the evaluated datasets, but do not establish a universal ordering of convolutional and graph architectures.

Fig. 7 illustrates an overview of the discussed modeling and evaluation approaches of this chapter. The classical machine learning approaches such as SVM, random forest, KNN, and LDA algorithms apply the extracted features for emotion classification. On the other hand, deep learning approaches including CNN, RNN, LSTM and attention-based models learn their representation either from EEG signals or the extracted features [22, 23]. In order to deal with individual variation and acquisition conditions, transfer learning leverages domain adaptation, domain generalization, and fine-tuning by utilizing discrepancy minimization, adversarial training, clustering and source-free methods [94, 96]. As shown in Fig. 7, the performance of the classifier is measured in terms of accuracy, precision, F1-score, ROC, AUC, and confusion matrix.

Model assessment should account for the intended prediction setting and the available supervision. Accuracy, precision, recall, F1 score, confusion matrices, and ROC based measures capture different aspects of classification behavior, but their meaning depends on the emotion classes, evaluation partitions, and averaging procedures. Egger et al. [118] reviewed emotion recognition across several physiological and behavioral modalities and noted that reported performance depends on the recognition task, features, classifiers, and dataset quality. This is a comparison of the literature, not a controlled experiment establishing that a particular EEG architecture is superior. Likewise, the attention and transfer experiments reported by Li et al. [119] must be interpreted within their own subject and session evaluation protocols.

Overall, the original studies support considering representation capacity, preprocessing, subject variability, and validation design together. The most defensible choice of a deep architecture is determined by the information represented in the input and by comparative experiments using consistent data partitions, rather than by a general claim that one architectural family is always preferable.

3.3.2.1 | Standalone Models. Standalone models are considered here as systems organized around one principal neural encoder and an emotion prediction head, without a separate target adaptation network or explicit fusion of several independent encoders. This is an architectural distinction rather than an assurance of improved domain generalization. A single neural encoder can still receive engineered EEG features, employ multiple training losses, or require participant specific calibration. In their systematic review, Roy et al. [23] documented the use of convolutional and recurrent networks across EEG analysis tasks, together with marked differences in input preparation and reported experimental methods.

Convolutional processing can learn localized patterns in EEG inputs represented as time series or structured feature arrays. A generic convolutional layer can be written as

$$\mathbf{H}^{(l)} = \sigma\left(\mathbf{W}^{(l)} * \mathbf{H}^{(l-1)} + \mathbf{b}^{(l)}\right), \tag{13}$$

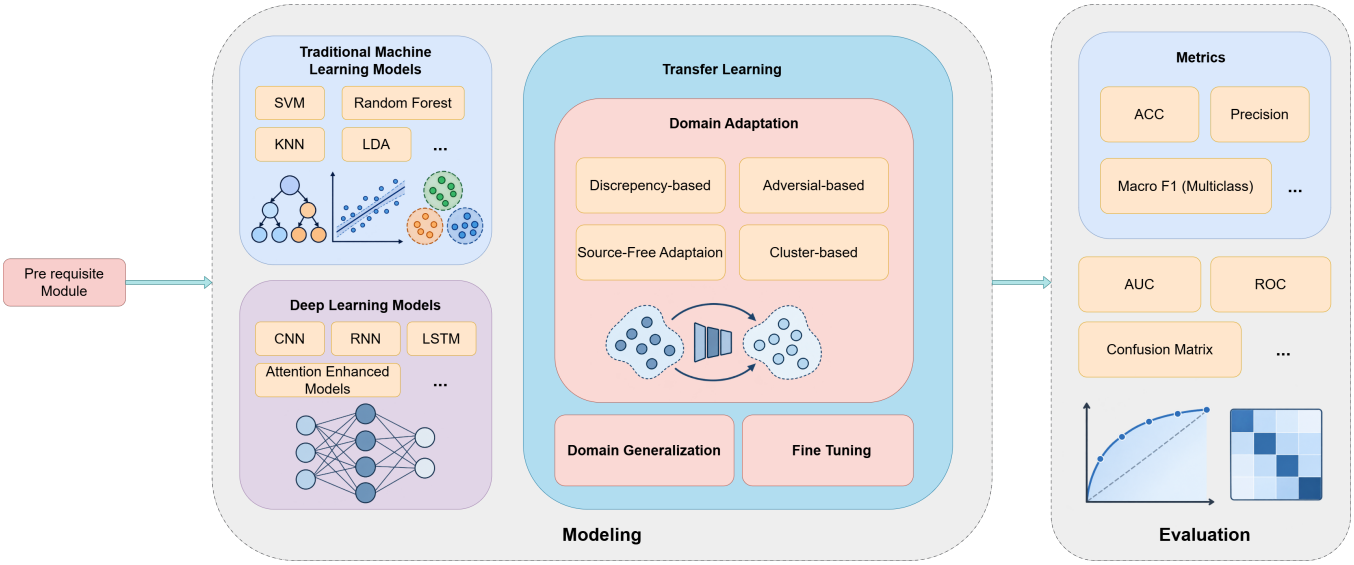


FIGURE 7 | Overview of modeling and evaluation approaches in EEG based emotion recognition, illustrating traditional machine learning, deep learning, transfer learning strategies, and commonly employed performance evaluation measures.

where $\mathbf{H}^{(l-1)}$ and $\mathbf{H}^{(l)}$ are the input and output representations, $\mathbf{W}^{(l)}$ and $\mathbf{b}^{(l)}$ are learnable convolutional parameters, $*$ denotes convolution, and σ is an activation function. This is a generic operation, not an exact reproduction of the network defined in an individual study. Li et al. [7] demonstrated one efficient CNN design, reporting that its proposed convolutional block required fewer parameters than the conventional block used for comparison. Their supervised contrastive component was also evaluated through comparisons with a classification loss alone on DEAP and DREAMER. Thus, the paper supports a specific efficiency and objective design, not a claim that every CNN is computationally preferable to recurrent alternatives.

Recurrent processing provides a different mechanism for combining observations across a sequence. Aydin [120] extracted phase space trajectory representations from six second EEG segments, applied principal component analysis to characterize phase domain complexity, and compared these descriptors with phase locking value information using an LSTM based learning approach. The study investigated EEG responses to nine elicited emotional conditions and comparisons between participant groups. Because its network relies on constructed phase domain descriptors, it should not be presented as an example of learning every representation directly from raw EEG or as proof of transfer to unseen participants.

It is also necessary to distinguish neural model design from conventional classifier adaptation. Wankhade and Doye [115] proposed an improved nearest neighbor classifier with preprocessing, feature extraction, and selection, evaluating it using SEED. Their IKNN approach is a conventional machine learning pipeline rather than a standalone deep model. It can therefore serve as a contrasting baseline category, but its results do not establish a memory or latency advantage for deep learning architectures. Similarly, the review by Egger et al. [118] compares physiological emotion recognition approaches across application settings, rather than experimentally benchmarking standalone CNN and RNN encoders under identical EEG conditions.

In summary, standalone encoders provide a relatively direct setting for studying how a specified representation and neural architecture contribute to emotion classification. Their suitability for deployment, limited training data, or unseen participants must still be established through measurements of recognition performance, computational requirements, and the relevant transfer protocol.

3.3.2.2 | Attention Enhanced Models. Attention mechanisms assign learned weights to components of an input representation and thereby modify the information available to later network layers. In EEG emotion recognition, these components may correspond to electrode channels, temporal positions, or encoded features. Attention is therefore best described by its target representation, weighting operation, and position within the network, rather than by a general assumption that it removes artifacts or identifies causal neural mechanisms.

Tao et al. [121] proposed an attention based convolutional recurrent neural network (ACRNN) that first applies channel attention to EEG inputs, uses convolution to extract encoded spatial information, and then integrates extended self attention into recurrent processing to model temporal information. The authors evaluated the framework on DEAP and DREAMER and compared it with the methods included in their experiments. Their design supports the relevance of learning channel and temporal weights jointly, but the experiments do not demonstrate that the attention coefficients themselves measure causal importance of a particular brain region.

A conventional scaled dot product attention operation is

$$\mathrm{Attn}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\sqrt{d_k}}\right)\mathbf{V}, \tag{14}$$

where $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are query, key, and value representations and d_k is the query and key dimension. This equation defines a general attention mechanism and is not presented as the exact channel attention implementation of Tao et al. A simple illustrative channel weighting operation is $\widetilde{\mathbf{X}} = \mathrm{diag}(\boldsymbol{\alpha})\mathbf{X}$, where $\boldsymbol{\alpha}$

contains learned channel weights. The form and normalization of these weights depend on the particular architecture.

Li et al. [119] developed the CADD-DCCNN framework, combining multiview attention with dilated causal convolution and a domain discriminator. The model treats channel specific differential entropy representations as separate views and uses attention to examine informative channels and time intervals. It was evaluated on SEED and DEAP, with an ablation analysis of the attention, convolution, and adaptation components. This study addresses both representation learning and distribution shift, so its reported improvements should not be attributed to attention alone. In particular, attention maps and changes in classification after removing selected channels provide evidence about model behavior under the evaluated conditions, not independent physiological validation of the highlighted electrodes.

These studies show that channel and temporal weighting can be integrated into EEG emotion recognition architectures and studied through ablation experiments. They do not establish that attention always improves unseen subject performance, suppresses ocular contamination, or reduces the number of physical electrodes required. Such claims would require direct comparisons with appropriate controls and acquisition conditions.

3.3.2.3 | Hybrid Models. Hybrid architectures combine processing mechanisms that emphasize different aspects of EEG structure. Rather than assuming that one operator captures all relevant activity, these models integrate convolutional processing, temporal sequence modeling, attention, graph operations, or representation learning objectives. The rationale is methodological complementarity; improved transfer across people or sessions remains a separate experimental question [35, 113].

One form of hybridization combines spatial or spectral feature extraction with recurrent modeling. Tao et al. [121] integrated convolutional operations with recurrent temporal processing and attention in ACRNN. Lew et al. [61] developed a regionally operated domain adversarial network that uses bidirectional gated recurrent units to represent EEG features from groups of channels and combines their representations with attention and domain discrimination. The latter authors investigated multiple evaluation protocols on DEAP and SEED-IV, illustrating that architecture comparisons depend on whether participants and trials are separated in training and testing.

An illustrative gated recurrent unit can be written as

$$\mathbf{z}_t = \sigma(\mathbf{W}_z\mathbf{h}_t + \mathbf{U}_z\mathbf{s}_{t-1} + \mathbf{b}_z), \tag{15}$$

$$\mathbf{r}_t = \sigma(\mathbf{W}_r\mathbf{h}_t + \mathbf{U}_r\mathbf{s}_{t-1} + \mathbf{b}_r), \tag{16}$$

$$\widetilde{\mathbf{s}}_t = \tanh(\mathbf{W}_h\mathbf{h}_t + \mathbf{U}_h(\mathbf{r}_t \odot \mathbf{s}_{t-1}) + \mathbf{b}_h), \tag{17}$$

$$\mathbf{s}_t = (\mathbf{1} - \mathbf{z}_t) \odot \mathbf{s}_{t-1} + \mathbf{z}_t \odot \widetilde{\mathbf{s}}_t, \tag{18}$$

where $\mathbf{h}_t$ is the input at time t, $\mathbf{s}_t$ is the recurrent state, and $\mathbf{z}_t$ and $\mathbf{r}_t$ are learned gates. These are generic GRU update equations, not a transcription of the full RODAN training objective.

Li et al. [67] proposed a multiview spectral spatial temporal masked autoencoder (MV-SSTMA) with a CNN and Transformer structure. The method first reconstructs randomly masked EEG channels during pretraining on unlabeled data, then calibrates the learned representation with a limited number of labeled recordings from the target participant. The authors evaluated emotion classification on SEED and SEED-IV and separately examined recognition when test inputs contained missing channels. Their results therefore support a particular combination of reconstruction pretraining and personalized calibration, not a claim of fully calibration free domain generalization.

Another hybrid approach combines attention, temporal convolution, and domain alignment. The CADD-DCCNN architecture of Li et al. [119] incorporates channel views, cross attention, dilated causal convolution, and an adversarial domain discriminator. Its experiments on SEED and DEAP investigate the contributions of these components to classification and transfer within the stated protocols. Such ablations are more informative for attributing gains to individual modules than an overall comparison of final accuracies, although they do not remove differences arising from datasets, preprocessing, or the availability of target domain information.

The broader reviews by Suhaimi et al. [35] and Li et al. [113] document the diversity of physiological representations and recognition workflows, but do not themselves demonstrate that combining more neural modules always improves EEG emotion recognition. Hybrid models introduce additional design choices concerning the representation passed between components, their training objectives, and their evaluation conditions. Their benefits should therefore be assessed through controlled ablation studies and consistent partitions, with particular attention to whether results concern individual calibration, unseen participants, or changes between datasets.

Overall, deep EEG emotion recognition encompasses standalone encoders, attention mechanisms, and hybrid learning systems with distinct assumptions about input structure and supervision. The available original studies establish the feasibility of these approaches within their evaluated settings, but do not support universal rankings based on model family alone. Transparent preprocessing, reporting of target data use, and consistent evaluation protocols remain necessary for meaningful conclusions about recognition accuracy and generalization.

4 | Datasets and Comparative Study

The selection of dataset affects the interpretation of the results of emotion recognition from EEGs due to differences in publicly available datasets in terms of the population, stimulation protocols, electrode placements, affective labels, and physiological modalities included in them. Thus, one should take into account the characteristics of the dataset along with the evaluation procedure, rather than considering accuracy results from different benchmarks as directly comparable. For instance, the ECNN-C method was tested on DEAP and DREAMER, yet the results for the two datasets are still conditional upon their respective task definitions and evaluation procedures [7]. The paper of Kukhilava et al. [5] shows considerable diversity of ground truth definition, metrics used to evaluate the performance, and split of data for EEG-based emotion recognition.

The datasets in Table 6 differ not only in scale but also in what their labels represent. DEAP, DREAMER and the first MAHNOB HCI experiment provide dimensional affect ratings, while SEED and SEED IV principally support recognition of experiment defined emotion categories. AMIGOS additionally includes basic emotion ratings and distinct individual and group viewing conditions, whereas ASCERTAIN links affective responses to personality measures [122, 6, 126, 41, 127]. MPED provides

TABLE 6 | Comparative overview of widely used EEG based emotion recognition datasets.

Dataset	No of Participants	Design	EEG Channel	Rate	Stimuli	Emotion targets	Annotations	Recorded modalities	Important qualification
DEAP [122]	32	One session	32	512 Hz recorded; 128 Hz processed	40 music video excerpts, 60 s each	Dimensional affect	Valence, arousal, dominance, liking and familiarity	EEG and peripheral signals; face video for 22 participants	Video is not available for all 32 participants
SEED [123, 29, 19]	15	Three sessions	62	1000 Hz recorded; 200 Hz processed	15 film clips per session, about 4 min each	Three emotions	Positive, neutral and negative target classes; postclip self assessment	EEG for 15; eye movements for 12	Sessions separated by approximately one week
SEED IV [124, 125]	15	Three sessions	62	200 Hz processed	72 clips in total; 24 per session	Four emotions	Happiness, sadness, fear and neutral	EEG and eye movements	Provided EEG features use nonoverlapping 4 s windows
DREAMER [6]	23	One experiment	14	128 Hz	18 audiovisual film excerpts	Dimensional affect	Valence, arousal and dominance, rated 1 to 5	EEG and ECG	Wearable EEG and ECG acquisition
MAHNOB HCI [126]	27	Two experiments; first for affect	32	256 Hz	20 emotional videos in first experiment	Dimensional affect and emotional keywords	Valence, arousal, dominance, predictability and keywords	EEG, peripheral physiology, face video, audio and gaze	Implicit tagging is a separate second experiment
AMIGOS [41]	40 / 37	Short video and long video experiments	14	128 Hz	16 short and 4 long videos	Dimensional affect and basic emotions	Valence, arousal, control, familiarity, liking and basic emotion ratings	EEG, ECG, GSR and video	40 participants in short video study; 37 in long video study
ASCERTAIN [127]	58	One experiment	1	32 Hz	36 video clips, 51 to 128 s	Dimensional affect and personality	Valence, arousal, engagement, liking, familiarity and Big Five traits	Single dry electrode EEG, ECG, GSR and facial measurements	One EEG sensor produces several derived information streams
MPED [128]	23	Two recording parts	62	1000 Hz recorded	28 selected video clips	Seven discrete categories	Six emotions and neutral; valence and arousal self ratings	EEG, ECG, GSR and respiration	Parts recorded with a rest interval of at least 24 h
FACED [129]	123	One experiment	32	250 or 1000 Hz recorded; 250 Hz processed	28 videos	Nine discrete categories	Target emotion labels and 12 self report rating items	EEG and behavioral ratings	Two acquisition cohorts used different original sampling rates
GAMEEMO [130]	28	Four game conditions	14	128 Hz	Four games, 5 min each	Valence and arousal quadrants	Self assessment manikin; four affective quadrants	EEG	Original data have 14 channels; single channel analyses are later model choices

seven categories together with multimodal physiological signals, and FACED includes nine categories and a larger participant cohort than the other datasets in the table [128, 129]. GAMEEMO differs from film based datasets because emotional responses were elicited through computer games and rated in valence and arousal dimensions [130].

Several numerical attributes require careful interpretation. The 200 Hz values for SEED and SEED IV describe distributed processed signals, while the original SEED recording rate is 1000 Hz. Similarly, the FACED acquisition comprised cohorts recorded at 250 Hz or 1000 Hz, with provided processed signals standardized to 250 Hz [123, 124, 129]. Furthermore, a study using a single channel from GAMEEMO should not cause the underlying dataset to be described as a single electrode acquisition. These

distinctions are relevant when evaluating spatial modeling, preprocessing requirements and the feasibility of reduced electrode deployment.

As described in Table 7, a data set is to be chosen based on the question being investigated. The multiple trials in SEED and SEED IV can be used to study temporal stability, while the large FACED sample can be used to study participant variability [19, 129]. Recordings from wearable EEG systems are useful for examining the constraints of consumer acquisition, but results from such datasets do not by themselves establish performance under unrestricted everyday movement or different electrode configurations [34]. Multimodal datasets permit comparisons between EEG and complementary signals when these streams and their labels are available for corresponding observations [14].

The comparison must also account for the evaluation unit. Kukhilava et al. [5] documented inconsistencies in ground truth, performance metrics and splitting strategies across EEG emotion recognition studies. Brookshire et al. [132] showed, in clinical EEG classification rather than emotion recognition, that assigning segments from the same participant to both training and testing can markedly overestimate performance on unseen people. This supports treating participant separation as a methodological requirement whenever the intended claim concerns unseen participants; it does not supply an EEG emotion recognition benchmark result. Across the reviewed datasets, reliable comparisons therefore require explicit reporting of recording rates, annotation targets, participant and session partitions, preprocessing scope and performance measures. No single dataset captures every acquisition or deployment condition, so reported accuracy should always be interpreted in relation to the dataset and protocol used to obtain it.

4.1 | Performance Metrics

Performance metrics provide a quantitative basis for assessing EEG based emotion recognition models in terms of classification accuracy, class specific performance, discrimination, and generalization. The choice is essential since there are variations among EEG datasets with regard to the class distribution, emotions and evaluation. In particular, Kukhilava et al. [5] pointed out some flaws in the performance metrics used, the ground truth definition and the data split in EEG emotion recognition works.

Accuracy, precision and recall are basic measures for classification problems that can be obtained from the confusion matrix. Accuracy measures the proportion of correctly predicted samples, while precision refers to the ratio of the correctly predicted positive samples and recall to the fraction of positive samples predicted [10]. In the case of binary classification problem, where TP, TN, FP and FN denote true positive, true negative, false positive and false negative, respectively, with $N = TP + TN + FP + FN$. These metrics are defined as:

$$\begin{aligned} \text{Accuracy} &= \frac{TP + TN}{N}, \\ \text{Precision} &= \frac{TP}{TP + FP}, \\ \text{Recall} &= \frac{TP}{TP + FN}. \end{aligned} \tag{19}$$

While accuracy is used for general measure of performance, it fails to reveal under-performance in recognizing emotions belonging to minority classes. In order to provide yet another measure of classification performance, F1 score is computed as the harmonic mean of precision and recall:

$$\text{F1} = \frac{2\,\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{20}$$

For multiclass emotion recognition, macro F1 assigns equal weight to the F1 score of each emotion category:

$$\text{MacroF1} = \frac{1}{C}\sum_{c=1}^{C} \text{F1}_c, \tag{21}$$

where C denotes the number of emotion classes. However, macro F1 metric does not assign any weights to the classes based on their occurrence rate, though classwise metrics still get affected by prediction mistakes as well as distribution of classes [5].

Balanced accuracy can serve as another metric for evaluating an imbalanced problem and places equal emphasis on sensitivity of all classes. For binary classification, it is defined as

$$\begin{aligned} \text{BalAcc} &= \frac{1}{2}(\text{TPR} + \text{TNR}) \\ &= \frac{1}{2}\left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP}\right). \end{aligned} \tag{22}$$

In multiclass classification, balanced accuracy is the average of recalls of all classes, $\text{BalAcc} = \frac{1}{C}\sum_{c=1}^{C} \text{Recall}_c$. The measure of macro F1 includes both precision and recall, while balanced accuracy is concerned only with class sensitivity. Chakladar and Chakraborty [80] provided both classical accuracy and balanced accuracy in their experiments on EEG-based emotion recognition.

The area under the receiver operating characteristic curve (ROC-AUC) is a metric of the separation capacity of the classifier. It can be defined mathematically as

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR})\, d\text{FPR}, \tag{23}$$

where $\text{FPR} = FP/(FP + TN)$ and $\text{TPR} = TP/(TP + FN)$. Unlike accuracy and F1, which are evaluated under a specified decision rule, ROC AUC summarizes discrimination across thresholds. For multiclass problems, the adopted averaging and class comparison strategy should be specified.

Beyond classification metrics, performance variability across participants and evaluation runs is important for examining recognition consistency. For $K > 1$ evaluation units with metric values m_k, the mean and sample standard deviation are calculated as

$$\overline{m} = \frac{1}{K}\sum_{k=1}^{K} m_k, \qquad s = \sqrt{\frac{1}{K-1}\sum_{k=1}^{K}(m_k - \overline{m})^2}. \tag{24}$$

$\overline{m} \pm s$ reporting gives the performance measure averaged and corresponding variation but interpretation of this measure highly depends on the nature of the evaluation units (whether they are related to participants, folds, sessions, repeated learning procedures etc.). Participant overlap was shown to cause overly positive estimate of performance when comparing EEG classification tasks in clinics by Brookshire et al. [132]. Thus, participant segregation is an important factor for generalization assessment.

TABLE 7 | Scenario driven selection of EEG emotion recognition datasets for evaluating cross subject generalization, cross session robustness, wearable deployment, multimodal fusion, dimensional affect prediction, and discrete emotion classification.

Experimental goal	Relevant datasets	Reason and evaluation qualification
Recognition in unseen participants	FACED [129]; SEED [29]; SEED IV [125]	FACED provides 123 participants for estimating variation among people. SEED and SEED IV have established participant identities and support evaluation with participants held out of training. Results still depend on the split and label scheme [131].
Recognition across recording sessions	SEED [19]; SEED IV [124]	Both datasets contain three sessions per participant, permitting explicitly separated session evaluation. Recording sessions must not be randomly mixed when claiming performance on unseen sessions.
Wearable acquisition and reduced electrode evaluation	DREAMER [6]; AMIGOS [41]; GAMEEMO [130]; ASCERTAIN [127]	DREAMER, AMIGOS and GAMEEMO use 14 electrode consumer systems; ASCERTAIN uses a single dry electrode. Later channel reduction experiments should be distinguished from the original recording configuration [34, 10].
Fusion of EEG with complementary modalities	DEAP [122]; MAHNOB HCI [126]; AMIGOS [41]; MPED [128]; ASCERTAIN [127]	Each provides EEG alongside at least one complementary physiological or behavioral modality. The available modalities and participant coverage must be checked before aligning or fusing observations.
Prediction of dimensional affect	DEAP [122]; DREAMER [6]; MAHNOB HCI [126]; AMIGOS [41]; ASCERTAIN [127]	These datasets provide self ratings of valence and arousal and, in some cases, other affective dimensions. Continuous regression and classification after discretization are different prediction tasks and require explicit reporting of label processing.
Discrete emotion categories	SEED [123]; SEED IV [124]; MPED [128]; FACED [129]; GAMEEMO [130]	SEED has three assigned emotion classes, SEED IV four, MPED seven and FACED nine. GAMEEMO can support four valence and arousal quadrants, which should not be confused with four basic discrete emotions.

The statistical tests can be additionally performed for comparisons of recognition algorithms. As for Shen et al. [26], they used the paired t test with Bonferroni correction, Song et al. [68] used the Wilcoxon signed rank test for their comparison experiments. It is worth selecting the appropriate statistical method and clearly specifying the units used for comparison.

In conclusion, evaluation of EEG emotion recognition task benefits from use of supplementary metrics alongside with accuracy. Besides the measures of class balance, macro F1 and balanced accuracy provide additional insight into class-specific performance, while ROC AUC measure assesses discriminative power of classifier on different thresholds.

4.2 | Experimental Analysis

Experimental evaluation of EEG emotion recognition systems depends on the relationship between training and testing participants. Subject dependent evaluation uses recordings from the same participants during model development and testing, whereas subject independent evaluation examines recognition performance on participants excluded from training. In Nath et al. [133], these approaches have been evaluated by means of DEAP database, and they found out that lower classification accuracies are obtained on subject independent setting for the classifiers that are studied. Moreover, transfer learning considers how knowledge can be transferred from source to target domain, as investigated in Li et al. [134]. In [103], Cimtay and Ekmekcioglu studied pretrained convolutional neural networks in different conditions and datasets, which shows the significance of classification evaluation in different perspectives.

Another challenge faced by EEG-based emotion recognition due to variability in participants and recording sessions is that distributions of features extracted from different sources might be unique. The approach taken by Chen et al. [135] in order to overcome this issue was MEERNet that integrates common and domain-specific feature extraction as well as distribution alignment across multiple sources. Likewise, the approach adopted by Cimtay and Ekmekcioglu [103] in solving the same problem is convolutional representation learning for recognition across different participants and datasets. In spite of that, their results are still relevant only for the respective datasets and experiments. Thus, the comparisons provided below keep their settings intact.

4.2.1 | Subject Dependent Evaluation

Subject dependent evaluation uses EEG from the same participant population in both model development and testing, typically through within subject partitioning or cross validation. This setting is useful for measuring the representational capacity of a model when subject specific calibration is available, and it generally produces higher accuracies than subject independent evaluation.

TABLE 8 | Comparative overview of representative subject-dependent EEG-based emotion recognition studies. Results are reported under their original experimental conditions. Differences in experimental protocols limit direct comparability across datasets

Study	Input feature(s)	Method	Dataset / evaluation	Accuracy (%)
Song et al. [92]	DE, PSD, DASM, RASM, and DCAU EEG descriptors represented over electrode graphs	DGCNN with dynamically learned inter electrode adjacency relationships	SEED; subject dependent	90.40 ± 8.49
Deng et al. [21]	Multichannel EEG reorganized into symmetry aware spatial representations preserving electrode topology	SFE-Net with symmetrical spatial feature extraction and convolutional representation learning	SEED; subject dependent	99.19
Jia et al. [69]	Three dimensional differential entropy representations encoding spatial, spectral, and temporal information	SST-EmotionNet using spatial, spectral and temporal attention with a 3D dense architecture	SEED and SEED-IV; subject dependent	SEED: 96.02 ± 2.17; SEED-IV: 84.92 ± 6.66
Guo et al. [136]	Differential entropy features extracted from the principal EEG frequency bands	DCoT combining depthwise convolutional representation learning with a Transformer encoder	SEED; subject dependent	93.83
Ding et al. [137]	DE features extracted by STFT from four frequency bands for DEAP and five bands for SEED; EEG electrodes form graph nodes	DGAT with a dynamically updated adjacency matrix and multi head graph attention for adaptive inter channel relationship learning	DEAP and SEED; five fold subject dependent cross validation	SEED: 94.00 ± 2.60; DEAP-V: 93.55 ± 3.89; DEAP-A: 93.19 ± 3.65
Xiong et al. [138]	DE, PSD, EVI, and ASI features from four frontal channels (FP1, FP2, AF3, AF4), mapped into pseudo RGB feature matrices	CNN-KAN-F2CA integrating convolutional feature extraction, FFT based frequency channel attention, and a Kolmogorov and Arnold Network classifier	SEED and DEAP; single subject subject dependent evaluation	SEED three class: 97.45; DEAP-V: 90.03; DEAP-A: 91.33; DEAP VA four class: 89.16
Lin et al. [139]	Bandpass filtered and baseline corrected multichannel EEG transformed into time and frequency representations	PconvCapsNet using selective partial convolution with synchronized channel bypass and primary and dense capsules with dynamic routing	DEAP and DREAMER; subject dependent	DEAP: A 98.03, V 97.81, D 98.11; DREAMER: A 97.20, V 96.49, D 97.04
Soundariya and Thangaraj [140]	Multiscale EEG components obtained through hierarchical wavelet packet decomposition and EWT with Kalman filtering	AWT-Net combining multiscale wavelet analysis, multi-head self-attention, a hybrid spatio temporal Transformer, and adaptive focal loss	DEAP; five fold subject dependent evaluation	99.61 window level; 99.30 trial wise grouped
Hua et al. [141]	Continuous temporal EEG for the spatio temporal branch and spectral spatial time and frequency maps preserving electrode topology for the second branch	BRAD-Net with brain region constrained self attention, CNN and Transformer temporal modeling, multi scale residual spectral spatial learning, and feature fusion	DEAP and DREAMER; subject dependent 10 fold cross validation	DEAP: V 97.44 ± 1.19, A 97.70 ± 1.13, D 97.97 ± 1.14; DREAMER: V 99.66 ± 0.22, A 99.78 ± 0.14, D 99.80 ± 0.16
Liu et al. [142]	Four dimensional dense EEG representation preserving channel topology together with temporal and multidimensional feature structure	TaFormer using triplet attention to learn cross dimensional, global spatial, and long range temporal information, followed by dual channel dynamic feature fusion	DEAP, SEED, DREAMER, and HBUED; subject dependent	DEAP: 98.88; SEED: 97.66; DREAMER: 89.55; HBUED: 99.85
Liu et al. [143]	Raw multichannel EEG signals processed directly without a separate handcrafted feature extraction stage	MLF-CapsNet combining multi level convolutional feature maps, primary capsules, dynamic routing, and a bottleneck layer for efficient end to end representation learning	DEAP and DREAMER; subject dependent emotion recognition	DEAP: V 97.97, A 98.31, D 98.32; DREAMER: V 94.59, A 95.26, D 95.13
Bao et al. [144]	Five band DE features extracted from segmented multichannel EEG; electrodes represented as graph nodes with connectivity initialized using channel relationships	MDGCN-SRCNN combining multi layer dynamical graph convolution with a style based recalibration CNN for joint shallow graph and deep convolutional feature learning	SEED and SEED-IV; subject dependent trial based evaluation	SEED: 95.08 ± 6.12; SEED-IV: 85.52 ± 11.58

Continued on next page

TABLE 8 | Comparative overview of representative subject dependent EEG-based emotion recognition studies.

Study	Input feature(s)	Method	Dataset / evaluation	Accuracy (%)
Shawky et al. [145]	Three dimensional representations constructed directly from multichannel EEG, with data augmentation used to enlarge the training set	3D-CNN designed to jointly learn spatial and temporal EEG characteristics from the constructed volumetric representation	DEAP; subject dependent binary valence and arousal classification	V: 87.44; A: 88.49
Xiao et al. [146]	Raw EEG transformed into four dimensional spatial, spectral and temporal representations	4D-ANN combining CNN based spatial and spectral processing with spatial and spectral attention and a self attention enhanced BiLSTM for temporal dependencies	DEAP, SEED, and SEED-IV; intra subject evaluation with five fold cross validation	DEAP-V: 96.90 ± 1.65; DEAP-A: 97.39 ± 1.75; SEED: 96.25 ± 1.86; SEED-IV: 86.77 ± 7.29
Gong et al. [147]	Spatial, spectral and temporal EEG representations with attention masks applied to spatial and spectral information at each temporal slice	ACTNN combining lightweight spatial and spectral attention, convolutional local feature extraction, and Transformer based temporal encoding with multi head self attention	SEED and SEED-IV; subject dependent evaluation	SEED: 98.47; SEED-IV: 91.90
Chen et al. [148]	Differential entropy features represented as EEG graphs, with electrodes treated as graph nodes and frequency band information incorporated into node representations	DAMGCN combining graph convolution with Transformer style dual attention over electrode channels and EEG frequency bands	DEAP, SEED, and SEED-IV; 10 fold subject dependent cross validation	DEAP: V 96.96 ± 2.24, A 97.17 ± 2.35, D 97.50 ± 2.03; SEED: 99.42 ± 0.24; SEED-IV: 96.86 ± 1.33
Liu et al. [149]	Differential entropy features from five EEG frequency bands reorganized into three dimensional representations according to electrode topology	DSSA Net integrating vertical and horizontal positional attention, spectral attention, and Transformer based temporal attention	DEAP, SEED, and SEED-IV; subject dependent evaluation	DEAP-V: 94.97 ± 4.23; DEAP-A: 94.73 ± 3.27; SEED: 96.61 ± 5.39; SEED-IV: 85.07 ± 11.93
Chen et al. [42]	DE, symmetric difference, and symmetric quotient features extracted under four EEG frequency bands and mapped to electrode topology brain maps	MPICNN with parallel convolutional inputs that jointly learn multiple frequency spatial brain map representations	DEAP and SEED-IV; subject dependent four class emotion recognition	DEAP VA four class: 98.71; SEED-IV four class: 92.55
Huang and Deng [150]	Complementary DE and PSD features from θ, α, β, and γ bands organized into four dimensional spatial, spectral and temporal representations	4D-MRSimNet combining SimAM enhanced multi resolution CNNs for spatial and spectral learning with BiGRU and temporal attention	DEAP and DREAMER; subject dependent valence and arousal recognition	DEAP: V 97.68 ± 1.72, A 97.61 ± 2.20; DREAMER: V 99.60 ± 0.87, A 99.46 ± 0.72
Wei et al. [151]	Raw multichannel EEG divided into non overlapping temporal windows and represented as patches containing local and global EEG information	TC-Net combining an EEG Transformer, EEG-PatchMerging for local feature preservation, and an emotion capsule module for modeling relationships among multidimensional features	DEAP and DREAMER; subject dependent evaluation	DEAP: V 98.76, A 98.81, D 98.82; DREAMER: V 98.59, A 98.61, D 98.67
Lu et al. [152]	Spatial and temporal EEG representations together with spatial and frequency representations derived from PSD and DE features across five frequency bands	Bi-ViTNet using parallel Transformer branches to learn spatial and temporal and spatial and frequency information followed by feature level fusion	SEED and SEED-IV; subject dependent 7:3 train and test evaluation	SEED: 97.55 ± 1.58; SEED-IV: 88.08 ± 6.32
Qian et al. [153]	Multichannel EEG represented as an adaptive graph in which electrode relationships are learned rather than fixed a priori	AGCN-SAT combining adaptive graph convolution, spatial attention, and Transformer based global context modeling	SEED; subject dependent emotion recognition	92.76 ± 6.16
Huang et al. [154]	Multichannel EEG represented through learnable spatial and temporal adjacency structures for dynamic modeling of channel connectivity and temporal dependencies	AST-CLNet integrating dual branch adaptive spatio temporal graph convolution, channel attention, local and global feature fusion, and supervised contrastive learning	SEED, SEED-IV, and DEAP; subject dependent/intra subject evaluation	SEED: 99.71 ± 0.45; SEED-IV: 97.91 ± 1.49; DEAP-V: 98.17 ± 3.91; DEAP-A: 95.92 ± 4.81

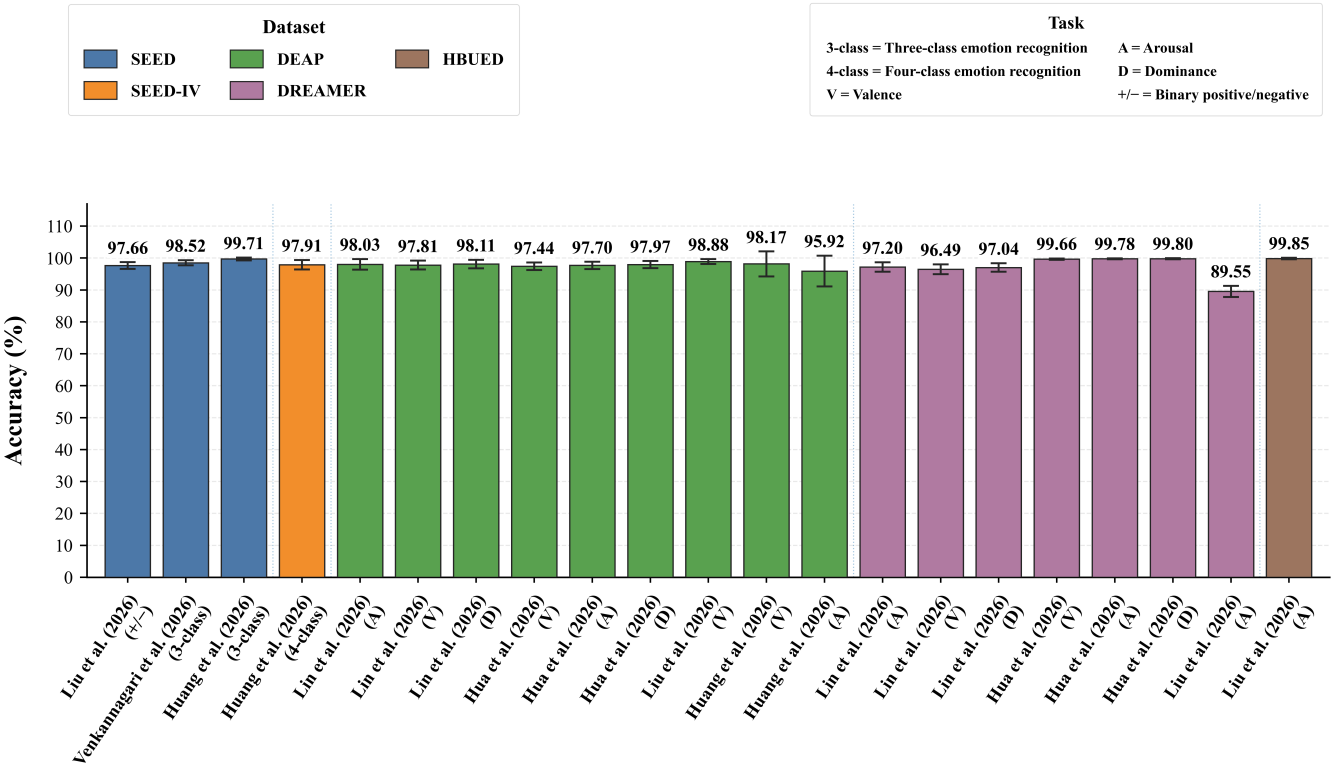


FIGURE 8 | Subject-dependent EEG emotion recognition accuracy across datasets, with error bars indicating reported standard deviations. Differences in experimental protocols limit direct comparability across datasets.

The progression in subject dependent EEG emotion recognition has been extensively investigated using conventional feature representations and increasingly sophisticated deep learning architectures. The comparative overview of representative works is given in Table 8. In earlier works, EEG features were mostly handcrafted, such as differential entropy (DE), power spectral density (PSD), and spatial EEG features, used together with convolutional or graph neural network models [138]. The most recent studies use adaptive graph convolution, attention, Transformers, and multi branch feature fusion to account for complicated spatial, spectral, and temporal dependencies involved in emotional response processing [137]. This indicates that there is an inclination to learn complementary representations of EEG data from neurons rather than focusing on specific frequency bands or electrodes.

From Table 8, it is observed that many studies have attained high accuracies for subject dependent evaluation on popular databases such as DEAP, SEED, SEED-IV, and DREAMER. Nonetheless, the reported results show significant heterogeneity of methodologies applied, including various configurations of emotion labels, input features, electrode placement, data segmentation, and validation strategy. Specifically, while some studies perform evaluation on individual EEG windows, others use trial wise grouping or other cross validation schemes. This is an essential issue since similar EEG segments presented in both training and testing datasets can provide overestimated accuracy. In addition, the difference between binary valence arousal classification, three class emotion recognition, and emotion categories make direct comparison of reported accuracies difficult. Therefore, the results presented in Table 8 should be treated as specific to the experiment conducted and cannot be viewed as ranking of classification architectures. Nonetheless, the results show the ability of deep learning to detect emotion related EEG patterns in individuals.

Subject dependent performance of EEG based emotion recognition has been visualized comparatively in Fig. 8 for six recent studies conducted on five standard datasets, i.e., SEED, SEED-IV, DEAP, DREAMER, and HBUED. The classification accuracy has been illustrated for various deep learning models used in those experiments. The experiments include binary classification of affective dimensions, multi class classification of emotions, and either window level or trial level classification of the data. The result shows that recently proposed subject dependent methods often have very high classification accuracy, which proves the efficiency of learning specific EEG spatial, spectral, and temporal representations for subjects.

4.2.2 | Subject Independent Evaluation

In the subject independent approach, the evaluation does not involve the training of the model using the target subject's data. The popular approach to do this is either leave one subject out or something similar. Thus, the subject independent approach checks if the learned features of the EEGs contain properties which reflect emotional states and are applicable to all subjects and do not depend upon the individual participant.

TABLE 9 | Comparative overview of representative subject-independent EEG-based emotion recognition studies. A = arousal, V = valence, and D = dominance.

Study	Input feature(s)	Method	Dataset / evaluation	Accuracy (%)
Song et al. [92]	DE, PSD, DASM, RASM, and DCAU representations associated with EEG channels	DGCNN with dynamically learned graph adjacency for modeling inter channel relationships	SEED; subject independent	79.95 ± 9.02
Li et al. [155]	Electrode level spatial and time and frequency EEG representations	STGATE combining spatial and temporal graph attention with a Transformer encoder	SEED, SEED-IV, and DREAMER; subject independent	SEED: 90.37 ± 6.18; SEED-IV: 76.43 ± 5.01; DREAMER-V: 77.44 ± 8.40; DREAMER-A: 75.26 ± 9.71
Chang et al. [156]	Event related EEG representations organized at emotion and stimulus scales	MSHCL using emotion scale and stimulus scale contrastive objectives in hierarchical hyperbolic space	SEED, MPED, and FACED; cross subject	SEED (3 class): 89.30 ± 4.40; MPED (7 class): 38.80 ± 7.10; FACED: 77.00 ± 3.60 (binary), 45.70 ± 3.40 (9 class)
Song et al. [157]	DE features from five frequency bands represented through channel and frequency graph structures	IAG with sample dependent graph construction, multi level graph convolution, graph coarsening, and recurrent modeling	SEED; subject independent	86.30 ± 6.91
Liu et al. [158]	Dynamic differential entropy obtained using empirical mode decomposition and differential entropy	DECNN integrating dynamic empirical decomposition with convolutional representation learning	SEED; leave-one-subject-out; binary positive/negative	97.56
Li et al. [159]	Time and frequency EEG channel features used to construct sample dependent electrode graphs	SOGNN with dynamically self organized graph construction and graph convolution for cross subject learning	SEED and SEED-IV; cross subject	SEED: 86.81 ± 5.79; SEED-IV: 75.27 ± 8.19
Priyasad et al. [160]	Multichannel EEG processed with independent channel wise representation encoders	Channel wise encoders coupled with geometric graph learning, multi head attention, and multi channel feature fusion	DEAP and DREAMER; cross subject	DEAP: V 69.43, A 69.72, D 70.72; DREAMER: V 64.98, A 63.71, D 61.81
Gupta et al. [75]	Channel specific EEG subbands obtained using flexible analytic wavelet transform, followed by information potential features	FAWT based subband decomposition, information potential feature extraction, feature smoothing, and random forest/SVM classification	SEED and DEAP; cross subject	SEED (3 class): 90.48; DEAP-V: 79.99; DEAP-A: 79.95; DEAP VA four class: 71.43
Huang et al. [161]	Three EEG feature matrices constructed to represent baseline adjusted activity and asymmetric responses of the left and right cerebral hemispheres	BiDCNN with a three input CNN architecture for learning inter channel and inter hemispheric differences	DEAP; subject independent, disjoint training and testing subjects	V: 68.14; A: 63.94
Fdez et al. [47]	Multitaper spectral EEG features with participant based normalization designed to suppress inter subject variability	Feed forward neural network with stratified normalization applied instead of conventional batch normalization	SEED; cross subject evaluation	91.60 (binary positive/negative); 79.60 (3 class positive/neutral/negative)
Wang et al. [56]	PSD features extracted from theta, slow alpha, alpha, beta, and gamma frequency bands and organized according to EEG brain regions	HSLT using electrode level and brain region level Transformer modules to learn hierarchical spatial dependencies	DEAP and MAHNOB-HCI; leave one subject out	DEAP-A: 65.75 ± 8.51; DEAP-V: 66.51 ± 8.53; MAHNOB-HCI-A: 66.20 ± 10.48; MAHNOB-HCI-V: 66.63 ± 6.51
Kumar et al. [162]	Nine bispectral quadratic phase coupling descriptors extracted from frontal EEG rhythms, followed by genetic algorithm feature selection	Explainable Boosting Machine using compact nonlinear bispectral features under nested LOSO evaluation	DEAP; nested leave one subject out	A: 70.39; V: 69.98
Li et al. [163]	DE features for SEED/SEED-IV and STFT based representations for MPED	GMSS combining graph representation learning with spatial jigsaw, frequency jigsaw, and contrastive self supervised auxiliary tasks	SEED, SEED-IV, and MPED; subject independent	SEED: 86.52 ± 6.22; SEED-IV: 73.48 ± 7.41; MPED: 28.49 ± 4.42

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TABLE 9 | Representative subject independent EEG emotion recognition results.

Study	Input feature(s)	Method	Dataset / evaluation	Accuracy (%)
Zhang et al. [164]	Multichannel frequency domain EEG features represented through sparsified learnable electrode graphs	SparseDGCNN using L_1 regularized graph connectivity and forward and backward splitting to suppress redundant connections	SEED, DEAP, DREAMER, and CMEED; subject independent	SEED (binary, gamma-band DE): 90.65 ± 10.93; DEAP-V: 64.66, DEAP-A: 66.65; DREAMER-V: 68.18, DREAMER-A: 67.12; CMEED-V: 66.83, CMEED-A: 75.47
Ji et al. [165]	MIMO denoised multichannel EEG represented across principal frequency bands	NR-BA-BFE-Net combining noise estimation, subspace filtering, graph based band feature extraction, and band wise self attention	SEED and SEED-IV; subject independent	SEED: 95.56 ± 4.18; SEED-IV: 83.15 ± 3.40
Pandey and Seeja [166]	Intrinsic mode functions obtained from EEG using variational mode decomposition	VMD based feature decomposition followed by a deep neural network for subject independent classification	DEAP; subject independent	V: 62.50; A: 61.25
Jana et al. [167]	Sparse spatio temporal EEG frames preserving electrode topology and temporal variations of multichannel signals	CapsNet trained on spatio temporal frame groups with leave one subject out and unseen subject evaluation	DEAP; cross subject	LOO subject mean: 57.17; unseen subject/unseen record mean: 51.07
Xu et al. [168]	Multichannel EEG sequences with temporal information retained from GRU units before subsequent spatial representation learning	GRU Conv combining GRU based temporal encoding with CNN based spatial feature extraction	SEED and DEAP; subject independent	SEED: 87.04; DEAP-A: 70.07; DEAP-V: 67.36
Yang et al. [169]	Ten linear and nonlinear EEG feature families including spectral, statistical, entropy, and Hjorth related descriptors	ST-SBSSVM integrating significance testing, sequential backward feature selection, and SVM classification	DEAP and SEED; cross subject	DEAP-V: 72.00; SEED: 89.00
Yin et al. [79]	Candidate EEG features represented through subject specific probability density functions to assess inter individual consistency	LRFS using probability density-based intersubject consistency, margin-based feature ranking, and ensemble classification across locally robust feature subsets	DEAP and MAHNOB-HCI; individual independent	DEAP-A: 65.10 (LSSVM); DEAP-V: 67.97 (LSSVM); MAHNOB-HCI-A: 67.43 (LSSVM); MAHNOB-HCI-V: 70.90 (KNN)
Li et al. [170]	Thirteen linear and nonlinear EEG features extracted using overlapping 4 s windows, followed by L_1 regularized feature selection	Ensemble of selected base classifiers with weights optimized by multiple objective particle swarm optimization	DEAP and SEED; leave one subject out	DEAP-A: 65.70; DEAP-V: 64.22; SEED: 84.44
Bai et al. [171]	PSD and DE frequency domain feature matrices upsampled using cubic spline interpolation to retain relationships across EEG channels	SECT combining a global channel Transformer with a shifted channel Transformer for localized brain region representation learning	SEED and DEAP; subject independent public dataset evaluation	SEED (3 class): 85.43; DEAP-V: 66.83; DEAP-A: 65.31
Su et al. [172]	EEG feature representations projected into discriminative low dimensional spaces through paired synthesis and analysis dictionaries	GA-PDPL combining projection dictionary pair learning with genetic optimization of model parameters	SEED, MPED, and GAMEEMO; leave one subject out	SEED: 69.89 ± 14.39; MPED: 24.87 ± 5.83; GAMEEMO binary: 64.34 ± 6.44; GAMEEMO four class: 49.01 ± 8.46
Guo et al. [173]	Differential entropy features together with functional connectivity representations calculated using phase synchronization	FC-FAN using separate residual feature extractors, temporal feature grouped attention, functional connectivity grouped attention, and feature fusion	SEED; cross subject	92.12 ± 3.32

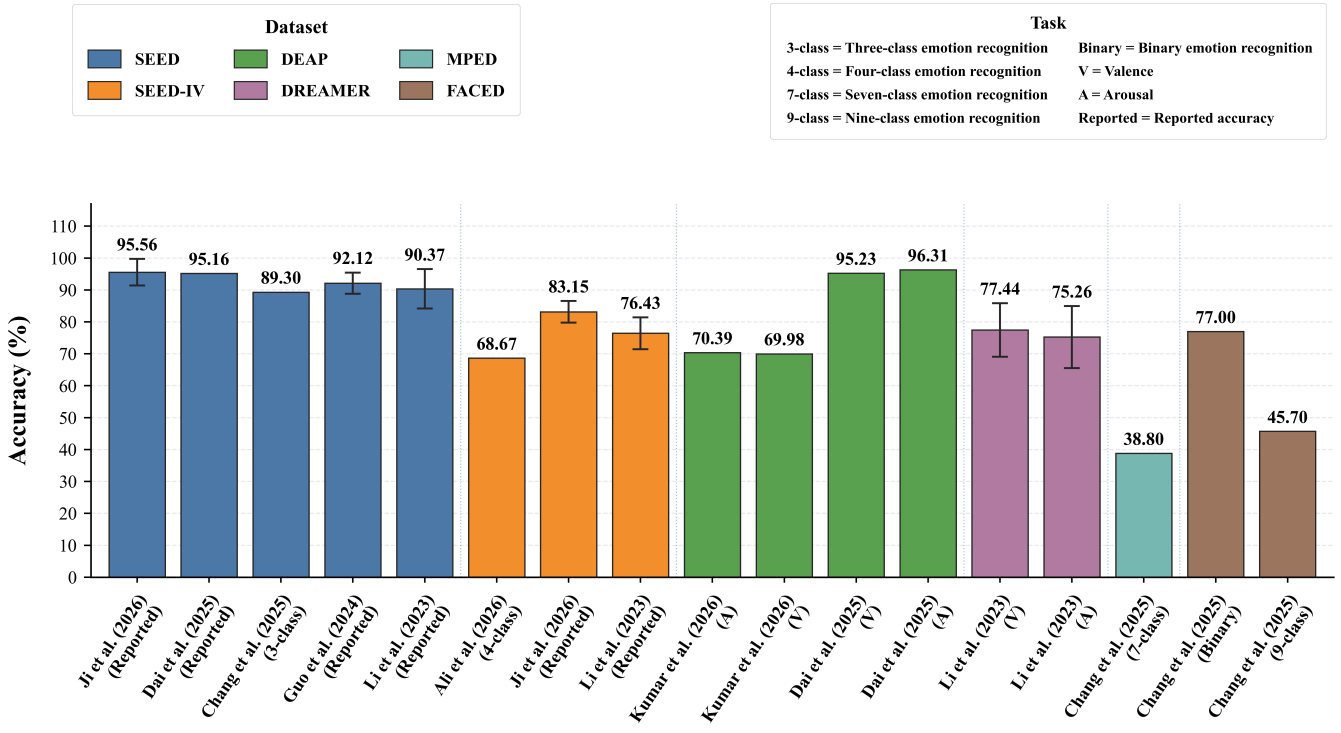


FIGURE 9 | Subject-independent EEG emotion recognition accuracy across datasets, with error bars indicating reported standard deviations. Differences in experimental protocols limit direct comparability across datasets.

Table 9 summarizes representative subject independent EEG emotion recognition studies spanning graph based learning, convolutional and recurrent architectures, Transformer based modeling, contrastive representation learning, and conventional feature selection methods. Unlike subject dependent evaluation, these protocols examine the ability of learned representations to recognize emotional states in participants excluded from model training. The reported results reveal considerable variation across architectures and datasets, indicating that recognition performance depends not only on feature representation and model complexity but also on the characteristics of the evaluation protocol.

Fig. 9 depicts the reported accuracy of six selected EEG based emotion recognition studies under subject independent or cross subject evaluation protocols. The comparison involves six datasets, namely SEED, SEED-IV, DEAP, DREAMER, MPED, and FACED, with binary affective classification and three , four , seven , or nine class emotion recognition tasks. In contrast to the subject dependent evaluation scenario, where the training and test data could come from the same subject, subject independent evaluation evaluates the capability of the model in recognizing emotions from unseen subjects. The latter is very important due to the fact that distributional variations of neural activity, emotional reactions, and EEG signals could be different across individuals. As shown in the figure below, the reported accuracy varies greatly among different datasets, classification tasks, and models.

Overall, the reviewed subject independent EEG emotion recognition studies employ representations of spatial, spectral, and temporal patterns to evaluate models on data from previously unseen participants. Nevertheless, the considerable variability in reported accuracy, including among studies using the same datasets, prevents the identification of a universally superior architecture. Reliable comparisons therefore require consistent subject disjoint partitions, clearly documented preprocessing procedures, comparable emotion label definitions, and evaluation using complementary metrics such as macro F1 and balanced accuracy. These considerations also emphasize the importance of investigating transfer learning and domain adaptation as complementary strategies for improving recognition robustness across previously unseen EEG domains.

4.2.3 | Transfer Learning

Transfer learning explicitly addresses distribution mismatch by transferring useful emotion related knowledge from a source domain to a target domain. In EEG emotion recognition, the domain boundary may correspond to a recording session, an individual participant, or an entirely different dataset. These scenarios can involve temporal, physiological, acquisition, and stimulus related discrepancies. The reviewed methods include feature level distribution alignment, multi source learning, calibration based adaptation, hybrid fine tuning, cross dataset unsupervised adaptation, and class conditional adversarial alignment.

The methods included in Table 10 illustrate the increasing diversity of transfer mechanisms. Adversarial and multi source approaches seek shared representations across subjects, while prototype based and pseudo labeling methods use class related structure to guide adaptation. More recent strategies incorporate Transformers, graph representations, curriculum based pseudo label refinement, and geometry aware alignment. Cross dataset

TABLE 10 | Representative transfer learning evaluations for EEG based emotion recognition. Results correspond to the original transfer setting of each method; cross subject, cross session, and cross dataset results should therefore be interpreted separately. A = arousal and V = valence.

Category	Study	Input feature(s)	Method	Dataset / transfer setting	Accuracy (%)
Adversarial feature transfer	Luo et al. [110]	Differential entropy EEG features	WGAN based domain adaptation for learning transferable source and target representations	DEAP and SEED; cross subject transfer	DEAP-V: 67.99; DEAP-A: 66.85; SEED: 87.07
Multi source domain adaptation	Chen et al. [174]	Five band differential entropy features from 62 EEG channels	MS-MDA, treating individual source subjects as separate domains and aligning marginal distributions with the target	SEED and SEED-IV; cross subject	SEED: 89.63 ± 6.79; SEED-IV: 59.34 ± 5.48
Hybrid DA and fine tuning	Lu et al. [112]	Differential entropy features extracted from δ, θ, α, β, and γ bands	DFF-Net combining an Emo-DA module with few shot target domain fine tuning	SEED and SEED-IV; cross subject	SEED: 93.37 ± 1.88; SEED-IV: 82.32 ± 5.38
Conditional adversarial transfer	Huang et al. [175]	DE and PSD features arranged as 14 s temporal sequences; each time step contains 310 features from 62 channels × five frequency bands	TT-CDAN using a temporal Transformer, MMD based domain adapter, and conditional domain discriminator	SEED and SEED-IV; cross subject	SEED: 93.62; SEED-IV: 82.16
Pseudo label propagation UDA	Zhong et al. [176]	Preprocessed emotion related EEG feature sequences from SEED, SEED-IV, and SEED-V, with temporal continuity exploited during propagation	DAPLP combining global distribution alignment, reliable pseudo label selection, graph based label propagation, correction, and smoothing	SEED, SEED-IV, and SEED-V; cross subject and cross session	Cross subject: SEED 89.44, SEED-IV 74.57, SEED-V 69.15; cross session: SEED 96.41, SEED-IV 82.20, SEED-V 84.70
Cross dataset UDA	Imtiaz and Khan [104]	Five band power spectral density features, DE was additionally evaluated in ablation experiments	Unsupervised domain adaptation with gradual proximity guided target data selection and confidence aware test time augmentation	SEED → DEAP and DEAP → SEED	SEED→DEAP: 59.68, DEAP→SEED: 67.44
Loop adversarial transfer	Yan et al. [177]	Electrode wise EEG feature matrices in which each sample contains n electrodes and d features per electrode	LATN integrating structure aware associative alignment, inner/outer product combination, and semi supervised pseudo label selection to suppress negative transfer	SEED, DEAP, and ECPL; cross subject	SEED (3 class): 94.34; DEAP-A: 89.21; DEAP-V: 76.12; ECPL (3 class): 96.33
Riemannian geometric transfer	Gao and Deng [178]	SPD covariance matrices computed from local DE feature vectors and represented on a Log Euclidean manifold	RGT-GAA combining a Log-Euclidean Graph Transformer with Fréchet mean geodesic alignment and adversarial domain adaptation	SEED and DEAP; LOSO cross subject, cross session, and one to one transfer	Cross subject: SEED 89.23; DEAP-V: 66.40; DEAP-A: 68.10; SEED cross session (many to one): 92.38 ± 3.47; SEED one to one cross subject: 74.56
Multi source dendrite adaptation	Liu et al. [179]	Five band DE features from δ, θ, α, β, and γ rhythms; 62 channel representations for SEED/SEED-IV and corresponding DEAP representations for cross dataset transfer	DD-MSDA using a dendrite-net common feature extractor, domain specific branches, source to target and target to target MMD alignment, and multiple domain specific classifiers	SEED, SEED-IV, and DEAP; cross subject, cross session, and cross dataset	Cross subject: SEED 87.57 ± 7.43, SEED-IV 74.79 ± 10.83; cross session: SEED 91.15 ± 7.81, SEED-IV 75.57 ± 11.71; cross dataset (six experiments): mean 57.66
Hierarchical adversarial transfer	Hong et al. [180]	Emotion evoked multichannel EEG representations from labeled source and unlabeled target subjects	HEAT with hierarchical global/class conditional adversarial alignment and curriculum based progressive pseudo labeling	SEED and SEED-IV; cross subject adaptation	SEED: 95.37 ± 5.17; SEED-IV: 82.50 ± 9.28

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TABLE 10 | Representative transfer learning evaluations for EEG-based emotion recognition.

Category	Study	Input feature(s)	Method	Dataset / transfer setting	Accuracy (%)
Multi source style transfer learning	Li et al. [98]	Five band DE features extracted from 1 s EEG segments	Multi source transfer with source selection, style transfer mapping between source and target EEG distributions, and source model integration, semi supervised variant exploits target test data transductively	SEED; calibrated cross subject transfer (supervised and transductive semi supervised settings)	MS-S-STM: 88.92; MS-Semi-STM (transductive): 91.31
Wasserstein multi source adaptation	Luo and Lu [181]	EEG feature representations extracted separately from multiple source subjects and the target subject	wMADA β with a shared feature extractor, source specific Wasserstein domain discriminators, and a common discriminator for multi source alignment	SEED, unsupervised cross subject adaptation	89.30 ± 4.00
Domain adaptive few shot learning	Ning et al. [182]	DE features from EEG channels spatially reorganized for convolutional representation learning	Few shot transfer combining deep feature extraction, MMD source and target alignment, attention mechanisms, and prototype based target classification	SEED and DEAP; 5 shot target adaptation (five labeled examples per class)	SEED: 96.51 ± 1.49; DEAP: 64.55 ± 6.65
Few label adversarial adaptation	Wang et al. [183]	EEG representations from selected source subjects plus a small labeled calibration subset from the target subject	FLADA with source selection, six group adversarial source and target alignment, shared feature extraction, and classifier aggregation	SEED and DEAP; few labeled target samples for cross subject adaptation	SEED: 89.32 ± 0.86; DEAP: 68.00 ± 0.05
Prototypical pairwise transfer	Zhou et al. [101]	EEG feature representations from SEED and SEED-IV used to construct class prototypes and pairwise relationships	PR-PL aligning source and target representations through prototypical learning and adaptive pseudo label based pairwise learning	SEED and SEED-IV; cross subject and cross session cross subject transfer	Session one cross subject: SEED 93.06 ± 5.12, SEED-IV 81.32 ± 8.53; cross session cross subject: SEED 85.56 ± 4.78, SEED-IV 74.92 ± 7.92
Pre trained cross dataset transfer	Wang et al. [184]	EEG representations from emotionally sensitive source subjects across SEED, SEED-IV, DEAP, and FACED	PESD pre trains a Vision Transformer encoder on emotion sensitive source data and transfers it using Mixup, triplet loss, and a domain discriminator	SEED, SEED-IV, DEAP, and FACED; cross dataset transfers with different source datasets	FACED→SEED: 93.14 ± 5.00; FACED→SEED-IV: 83.18 ± 24.58; FACED→DEAP: 93.53 ± 4.32; DEAP→FACED: 92.55 ± 6.86
Two discriminator adversarial transfer	Liu et al. [185]	310 D five band DE vectors (62 channels $\times$ 5 bands) extracted from 0.5 s EEG segments	TD-DANN with adaptive graph convolution, a source and target domain discriminator, and an individual discriminator for domain invariant and subject adaptive representations	SEED and SEED-IV; cross subject domain adversarial transfer	SEED: 89.45 ± 5.87; SEED-IV: 77.13 ± 7.97
Adaptive graph domain adaptation	Luo et al. [186]	DE features used to construct adaptive EEG graphs and dynamic frequency and spatial representations	AGCDAN with multi branch source/target feature extraction, discriminator free marginal alignment, subdomain metric learning for conditional alignment, and decision fusion	SEED, SEED-IV, and DEAP; multi source cross subject domain adaptation	SEED: 89.68; SEED-IV: 68.61; DEAP: 68.13
Flexible source aggregation	Ran et al. [187]	Multi source EEG representations dynamically aggregated according to source and target distribution similarity	FSA-Net with similarity based source aggregation, global MMD alignment, local MMD plus contrastive subdomain adaptation, and domain aware weighted ensemble prediction	SEED and SEED-IV; cross subject domain adaptation	SEED: 90.80; SEED-IV: 82.74

experiments, including transfers between SEED and DEAP, examine a particularly challenging setting because differences in stimulus protocols, acquisition procedures, and participant populations may coincide. Such studies broaden evaluation beyond the more common within dataset cross subject scenario, although their reported accuracies remain dependent on the exact task and adaptation protocol.

4.2.4 | Brain Region, Channels and Frequency Band Importance

The EEG correlates of emotion occur within more extensive neural networks through the interaction of various cortical sites, electrodes, and frequencies rather than through localized activity. Frontal, temporal, parietal, and occipital regions participate in wider systems linking emotional processing with cognition and perception. Subcortical structures, including the amygdala, hippocampus, thalamus, and hypothalamus, are also components of interacting cortical subcortical systems. Thus, the network account does not assign emotional states to anatomically isolated emotion specific centers [188].

Understanding these distributed processes is essential for EEG based emotion recognition because scalp EEG captures temporal, spectral, and spatial patterns associated with cortical neural activity. Nevertheless, its limited spatial specificity prevents reliable isolation of individual subcortical structures, requiring cautious anatomical interpretation [189]. Choosing relevant channels, brain regions, and frequency bands can contribute to better computation efficiency and physiological interpretation. But their relevance is highly individualistic, which means that using static channels and bands might not be sufficient, especially in terms of recognition of subjects-independent [190].

As summarized in Table 11, selected studies reported informative frontal, temporal, and centro parietal electrodes and frequently selected alpha , beta , and gamma band features in particular datasets. Other studies used adaptive graph construction, channel attention, brain region priors, and frequency aware representations to model spatial and spectral information. The reported regional and band related findings are study and protocol specific; they do not establish a universal electrode subset or frequency band ranking for EEG based emotion recognition.

5 | Discussion

The literature reviewed in this work shows that EEG based emotion recognition has progressed from conventional feature engineering pipelines toward increasingly sophisticated representation learning frameworks that model spatial, spectral, temporal, and inter channel dependencies. Nevertheless, improvements in reported classification accuracy do not translate directly into equivalent improvements in real world reliability. The central issue is that EEG emotion recognition is simultaneously a representation learning problem and a generalization problem: a model must identify affect related neural patterns while remaining robust to differences in participants, recording sessions, devices, experimental protocols, and environmental conditions. Recent systematic evidence likewise shows a persistent performance advantage for subject dependent evaluation over subject independent testing, indicating that individual variability remains a major obstacle to generalized affective decoding [194].

This distinction is particularly important when interpreting the comparative results summarized in Tables 8 and 9. Subject dependent models can exploit participant specific characteristics and therefore describe personalized recognition under the evaluated conditions. Subject independent evaluation imposes a different requirement because the test participant is excluded from model training, whereas transfer learning and domain adaptation may use source domain information and, depending on the protocol, labeled or unlabeled target domain data for adaptation. Consequently, reported accuracies should not be ranked without considering the associated subject partition, target data access, label configuration, session structure, and validation protocol. From this perspective, methodological progress should increasingly be judged not only by peak accuracy but also by robustness, calibration burden, computational feasibility, and the degree to which performance persists under realistic distribution shifts.

5.1 | Practical Implications and Real-World Applications

The practical importance of EEG emotion recognition arises from its capacity to introduce neurophysiological information into adaptive human computer and brain computer interaction. Unlike interfaces that rely exclusively on explicit commands, EEG informed systems can potentially infer changes in affective or cognitive state and use this information to modify system behavior. This capability is relevant to adaptive interfaces, assistive technologies, interactive environments, and multimodal affective systems in which EEG is combined with behavioral or physiological modalities. Reviews of EEG based BCI applications identify usability, portability, signal quality, and individual differences as important barriers to operation outside controlled settings [195].

The same technological foundation has implications for safety and learning environments. EEG has been investigated for fatigue detection during simulated driving. The E-Key framework evaluated personal identification and fatigue detection using wearable dry-sensor EEG recorded from 31 participants during a 90 min simulated driving task [196]. In educational environments, EEG based attention feedback has also been explored as a means of providing real time neurofeedback during learning tasks [197]. These examples should not be interpreted as evidence that emotion classifiers can be transferred directly to fatigue or attention without retraining. Rather, they illustrate a broader practical implication: advances in portable EEG acquisition, low latency preprocessing, representation learning, and individualized calibration developed for affective computing can contribute to a wider family of neuroadaptive systems.

Healthcare and personalized digital environments represent another important direction, although the clinical implications require especially cautious interpretation. EEG contains information relevant to functional brain organization, and a systematic review of EEG functional connectivity in depression reported differences in some frequency bands, while emphasizing substantial methodological heterogeneity and the inability to draw firm conclusions about the direction and anatomical location of these differences [3]. Virtual reality interventions have also been studied in mental health contexts; however, the cited review evaluated

TABLE 11 | Representative findings on brain region, channel, connectivity, and frequency band importance in EEG based emotion recognition.

Study	Method	Dataset / Setting	Brain Region and Channel Findings	Frequency Findings	Key Observation
Nakisa et al. [191]	Evolutionary feature and channel selection	DEAP, MAHNOB, mobile EEG	Electrode selection was nonuniform. The prominent DEAP channels identified through weighted electrode usage were Fp1, F7, FC5, AF4, CP6, PO4, O2, T7, and T8. The study also compared electrode selection patterns on MAHNOB and a five channel mobile EEG dataset.	Among frequency domain features, alpha and gamma band PSD features were selected frequently on DEAP; alpha , beta , and gamma band PSD features were frequently selected on MAHNOB.	The feature selection framework was evaluated on two public datasets and a five channel mobile recording dataset. Electrode and feature selection depended on the dataset.
Yang et al. [190]	Learnable channel weighting and core graph extraction	SEED	CWGCN learned channel importance parameters and extracted a core EEG network. CCSR GCN subsequently used the extracted information for emotion recognition.	The reported contribution concerns graph/channel selection, not a universal frequency band ranking.	The method used data driven channel importance to identify and utilize a task relevant EEG core network.
Li et al. [159]	Self organized graph neural network	SEED, SEED-IV; cross subject	Learned topographic maps assigned large weights to prefrontal and centro parietal electrodes, including F7, FP2, and CPz. The graph structure was dynamically constructed for each input rather than fixed in advance.	In the SEED single band experiment, theta yielded the highest SOGNN accuracy (76.00%); beta (72.54%) and gamma (71.70%) were also evaluated.	The all band model reported mean leave one subject out accuracies of 86.81 $\pm$ 5.79% on SEED and 75.27 $\pm$ 8.19% on SEED-IV.
Ma et al. [63]	Spatial frequency shifted window attention and temporal self attention	SEED, SEED-IV	Electrode locations were mapped into spatial representations; five frequency band feature maps formed the channels of an image processed with shifted window attention.	Five bands were jointly represented through differential entropy features; the reported results do not establish a single universally dominant band.	SFSWTS reported average accuracies of 99.46% on SEED and 95.41% on SEED-IV.
Valderrama and Sheoran [55]	Spatial and temporal attention	SEED, SEED-IV, SEED-V; subject independent	Channels identified as consistently relevant along the head circumference included Fp1/Fp2, F7/F8, FT7/FT8, T7/T8, TP7/TP8, P7/P8, PO7/PO8, and O1/O2.	The investigation primarily addressed channel relevance, rather than establishing a dominant spectral band.	Mean subject independent accuracies were 79.3%, 69.5%, and 60.7% for SEED, SEED-IV, and SEED-V, respectively.
Xu et al. [192]	Frequency aware self attention and uncertainty aware contrastive learning	SEED, DEAP, DREAMER, FACED	The proposed UACL-Net models EEG representations without reporting a universal optimal subset of scalp electrodes.	FASA applies attention to the real and imaginary parts of frequency domain features; it does not establish a fixed ranking of conventional EEG frequency bands.	The article reports accuracies of up to 94.88%, 98.71%, 96.91%, and 99.29% on SEED, DEAP, DREAMER, and FACED, respectively.
Hua et al. [141]	Brain region constrained attention	DEAP and DREAMER; subject dependent 10 fold CV; DREAMER LOSO additionally evaluated	Channels were partitioned by brain region; self attention in the spatio temporal branch was restricted to channels within each region.	A separate spectral spatial branch was fused with spatio temporal features; the article did not identify a universal best frequency band.	BRAD-Net combined within region attention with spectral spatial feature learning and evaluated subject dependent recognition on both datasets and subject independent recognition on DREAMER.

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TABLE 11 | Representative findings on brain region, channel, connectivity, and frequency band importance in EEG based emotion recognition (continued).

Study	Method	Dataset / Setting	Brain Region and Channel Findings	Frequency Findings	Key Observation
Lin et al. [193]	Brain region knowledge based dual stream Transformer	SEED, SEED-IV	Electrodes were organized into brain regions. The reported visualization identified temporal lobe channels as particularly important for recognition.	The two branches jointly modeled spatio temporal and spatio spectral features rather than selecting a single EEG frequency band.	The region based method was designed to reduce sensitivity to electrode shifts and reported 97.81% accuracy on SEED and 90.76% on SEED-IV.
Kannadasan et al. [83]	Differential evolution based feature selection with SVM	DEAP; subject independent evaluation reported by authors	A differential evolution algorithm selected a reduced EEG feature subset for emotion classification; specific regional dominance could not be independently checked against the accessible primary source text.	Band specific channel selections were not independently verifiable from the accessible original text; no universal band ranking is claimed here.	The original article reports 73.60% for valence and 74.23% for arousal; its abstract also reports 71.88% for dominance and 71.80% for liking.

VR interventions rather than EEG driven closed loop adaptation. Using EEG derived affective information to select or adapt immersive stimuli therefore remains a potential application, not a result established by that review [198]. EEG derived information has also been explored in recommendation systems, including neuro recommendation approaches that use brain responses to support personalized product selection [199]. Together with the expanding use of EEG in immersive virtual environments [200], these applications suggest that the long term value of emotion recognition may lie less in assigning an isolated emotion label and more in enabling adaptive systems that respond continuously to changes in the user's internal state.

The practical significance of the field therefore depends on a transition from benchmark oriented classification toward reliable affect aware interaction. Such a transition requires models that preserve performance across individuals and contexts, operate with acceptable computational latency, tolerate imperfect wearable recordings, and provide outputs whose uncertainty can be interpreted by downstream systems. High within dataset accuracy remains scientifically useful, but the more consequential question is whether an affective model can support a safe and meaningful decision when the user, recording environment, or acquisition hardware differs from the conditions under which the model was developed.

5.2 | Key Challenges and Future Research Directions

Despite substantial methodological progress, several limitations continue to restrict the generalizability and translational value of EEG emotion recognition. These limitations are interconnected rather than independent. Restricted datasets can increase overfitting risks, subject dependent results do not establish unseen subject performance, and inter subject variability can limit generalization or necessitate target specific calibration. Meanwhile, practical systems introduce additional sources of noise, latency, and user variability that are typically absent from benchmark experiments. A productive future research agenda should therefore address dataset design, personalized modeling, generalized modeling, and deployment requirements as components of a single validation pipeline rather than as separate research problems.

5.2.1 | Dataset Diversity, Annotation, and Standardization

Dataset design remains one of the most fundamental constraints on progress. Existing EEG emotion recognition research is concentrated around a relatively small number of benchmark datasets, and systematic reviews show that DEAP and SEED account for a disproportionately large share of experimental studies [194]. Although this concentration facilitates methodological comparison, it can also encourage optimization toward the statistical properties of a few established benchmarks. Many datasets were acquired using controlled laboratory elicitation, modest participant populations, fixed stimulus sets, specific electrode montages, and a limited number of emotion categories. Such conditions are valuable for experimental control but do not fully represent the variability expected in naturalistic affective states, where emotion intensity, mixed emotions, contextual factors, and spontaneous transitions are substantially more difficult to standardize.

A related challenge is heterogeneity across databases. Differences in sampling rate, electrode density, reference configuration, preprocessing, stimulus duration, segmentation strategy, emotional annotation, and class definition complicate direct cross dataset comparison. Reviews of automated EEG emotion recognition repeatedly identify variability in datasets, preprocessing, feature extraction, and evaluation methodology as barriers to reproducibility and broader generalization [12]. Future datasets should therefore place greater emphasis on transparent metadata, standardized reporting of acquisition and preprocessing parameters, subject wise split definitions, repeated session recordings, and demographically broader cohorts. Multi site and longitudinal acquisition could help assess whether observed patterns reflect emotions rather than participant-, session-, or device-specific variation.

Future dataset development should also reconsider how emotional ground truth is represented. Restricting affect to a small set of discrete labels or binary high/low dimensions simplifies classification but may conceal transitional, ambiguous, or mixed emotional states. More fine grained labels, confidence or intensity measures, repeated self assessment, and combinations of dimensional and categorical annotations could provide richer supervision. Naturalistic and immersive elicitation paradigms may further help bridge the gap between laboratory stimuli and real world affect, provided that the increased ecological validity is accompanied by careful control of movement, sensory load, and recording artifacts. The objective should therefore not be to maximize dataset size alone, but to increase the diversity of subjects, sessions, contexts, and emotional conditions in a way that meaningfully tests generalization.

5.2.2 | Subject-Specific Modeling and Personalization

Subject dependent modeling remains valuable because it can exploit electrophysiological characteristics that are specific to an individual. As shown in Table 8, many modern architectures obtain strong within subject performance by combining spatial topology, frequency domain information, temporal dependencies, graph representations, or attention mechanisms. These results show that the evaluated models can distinguish the reported emotion labels under their respective within-subject protocols. However, they primarily characterize personalized recognition and should not be interpreted as equivalent evidence of performance for previously unseen users.

A central methodological difficulty is that the quantity of data available for a single participant is usually limited relative to the capacity of contemporary deep networks. Increasing architectural complexity can therefore improve feature extraction while simultaneously increasing susceptibility to participant specific overfitting. Future subject specific systems should emphasize efficient personalization rather than unrestricted model complexity. Promising directions include compact hybrid architectures that integrate complementary spatial, spectral, and temporal representations, parameter efficient adaptation, representation pretraining followed by limited personalization, and explicit regularization of highly expressive attention or graph modules. More generally, systematic evidence suggests that multimodal and feature fusion strategies can provide complementary information, but the resulting models must be evaluated against their additional computational and data requirements rather than accuracy alone [201].

Personalized modeling should also incorporate temporal stability as an explicit evaluation criterion. A model trained and evaluated within one recording session may perform well while remaining sensitive to electrode placement, impedance, mood baseline, fatigue, or physiological changes on another day. Repeated session validation is therefore necessary to determine whether a "subject specific" model represents enduring affective characteristics or merely session specific signal structure. Future personalized systems would benefit from calibration procedures that update only the components required to accommodate session drift while retaining previously learned affective representations, thereby reducing the amount of new labeled EEG required for each use.

5.2.3 | Generalization Across Domains

Generalized emotion recognition is more difficult because EEG distributions can vary substantially between individuals. Differences in anatomy, scalp conductivity, electrode contact, baseline oscillatory activity, cognitive appraisal, emotional regulation, and stimulus interpretation can all alter the observed signal without necessarily implying a different affective state. Consequently, a feature that is discriminative within one participant may be unstable when applied to another. Reviews of EEG emotion recognition systems consistently identify this inter individual variability as a principal reason why subject independent performance remains below subject dependent performance across many benchmark settings. This generalization gap is particularly important because practical systems cannot usually assume that a fully personalized model and a large labeled calibration dataset will be available for every new user.

Future work should therefore distinguish clearly between different forms of generalization. Pure subject independent evaluation should retain complete separation between training and test participants and should not be conflated with algorithms that access target subject data during adaptation. Unsupervised domain adaptation may use unlabeled target data, semi supervised adaptation may use limited labeled calibration, and domain generalization methods attempt to operate without target domain access. Cross session and cross dataset experiments introduce additional levels of distribution shift. Reporting these settings separately, as in Tables 9 and 10, is essential because they answer different scientific questions.

The next stage of generalized modeling should move beyond optimizing a single benchmark split toward hierarchical stress testing. A model may first be evaluated using leave one subject out validation, then across sessions, acquisition devices, electrode subsets, and ultimately independent datasets. Transfer learning, domain adaptation, invariant representation learning, and source free adaptation can then be assessed according to how much target information they require and whether adaptation remains stable when source and target distributions differ substantially. Collaborative approaches such as federated or privacy preserving learning may also become valuable when multi institutional EEG data cannot be centralized. Importantly, generalized models should report not only overall accuracy but also class balanced measures, variability across held out participants, and failure cases, an improvement in mean performance has limited practical value if recognition becomes unreliable for a substantial subset of users.

5.2.4 | Real-World Deployment, Safety, and Ethical Considerations

Real world deployment introduces constraints that are largely absent from offline benchmark studies. Wearable EEG systems must contend with movement artifacts, changing electrode contact, environmental electrical noise, limited channel counts, setup time, user comfort, computational latency, and battery or hardware limitations. These factors interact directly with algorithm design: models that depend on dense electrode arrays, lengthy preprocessing, or repeated calibration may be unsuitable even if they achieve high laboratory accuracy. Reviews of EEG based BCI applications therefore emphasize portability and

practical usability as central requirements for broader adoption [195]. Immersive environments introduce additional challenges because head and body movement, visually complex stimuli, and the interaction between EEG hardware and head mounted displays can alter recording conditions, and the EEG VR literature identifies experimental design and technical integration as continuing areas for improvement [200].

Deployment also raises questions that cannot be resolved by classification performance alone. An affect sensitive system may influence recommendations, educational feedback, safety alerts, or therapeutic content, meaning that erroneous predictions can alter the user's experience or subsequent behavior. Models should therefore communicate uncertainty, avoid presenting probabilistic affect estimates as definitive psychological judgments, and include mechanisms for user control and disengagement. These concerns become especially important in mental health applications. A systematic review of VR interventions reported beneficial effects on stress and symptoms in many included studies, but effects on positive functioning were less consistent; the review did not evaluate EEG emotion classifiers as clinical decision systems [198]. EEG emotion recognition should accordingly be positioned as a complementary sensing technology within professionally validated clinical workflows rather than as an autonomous diagnostic authority.

Privacy and ethical governance will become increasingly important as affective BCIs move toward continuous or personalized use. EEG contains information beyond the specific emotion label targeted by a classifier, and persistent neural data therefore require strong safeguards for consent, storage, transmission, secondary use, and model updating. Application oriented research should incorporate these considerations during system design rather than after model development. Real time validation should also evaluate how systems behave when signals are noisy, when confidence is low, and when users intentionally disengage. These requirements favor models that are not merely accurate, but also computationally efficient, interpretable at an appropriate level, uncertainty aware, and capable of failing safely.

Overall, the future of EEG based emotion recognition depends less on incremental improvements to benchmark accuracy than on closing the gap between controlled experimental performance and dependable use across people, sessions, devices, and environments. Larger and more representative datasets, carefully separated evaluation protocols, efficient personalized models, stronger cross subject generalization, and application aware validation should therefore be pursued jointly. Progress along these dimensions would shift the field from demonstrating that emotional information can be decoded from EEG toward establishing when, for whom, and under what conditions such decoding is sufficiently robust to support meaningful real world decisions.

6 | Conclusion

The presented review offers an integrated analysis of emotion recognition based on EEG, uniting the psychological and neurophysiological background of emotions with the process of experimental design, EEG recording, signal processing, feature extraction, classification and evaluation. In contrast to method-focused reviews, this work discusses the impact of all these aspects and their interaction on the performance of emotion recognition, its reliability and generalization. Traditional approaches and deep learning techniques, data augmentation, transfer learning and domain adaptation are considered in terms of their importance for dealing with data scarcity and variability of different subjects. Special attention is paid to the difference between subject dependent recognition performance and generalization to new subjects and datasets. Reproducibility of experiments and evaluation methods are discussed, as well. Such a perspective allows understanding the existing trade offs in methodology and limitations of EEG based emotion recognition. Future work should be concentrated on the development of evaluation protocols, cross subject and cross dataset generalization and computationally efficient methods.

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